Mycobacterium tuberculosis Report

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SUMMARY

Field	Value
Sample name	ERR9030255
Species	Mycobacterium tuberculosis
Phylogenetic lineage	lineage1.1.2
Family	Indo-Oceanic (EAI)
Spoligotype Octal	477403377413771
Spoligotype Binary	1001111111000000110111111111000010111111111
Median Depth	118.0
% reads mapping	100.0
Drug resistance	Sensitive
Notes	-
Sublineage	lineage1.1.2
RD	RD239

SPOLIGOTYPE

Field	Value
Spoligotype_octal_code	477403377413771
Spoligotype_binary	1001111111000000110111111111000010111111111
Spoligotype SB Number	N/A

DRUG RESISTANCE SUMMARY

Predicted Drug Resistance

Drug	Confidence

None

Predicted Drug Susceptibility

Drug	Confidence
clofazimine	No mutation
delamanid	No mutation
linezolid	No mutation
ofloxacin	No mutation

Other / Unknown

Drug	Confidence
amikacin	Uncertain significance
bedaquiline	Uncertain significance
capreomycin	Uncertain significance
cycloserine	Uncertain significance
ethambutol	Uncertain significance
ethionamide	Uncertain significance
isoniazid	Uncertain significance
kanamycin	Uncertain significance
levofloxacin	Uncertain significance
moxifloxacin	Uncertain significance
para-aminosalicylic_acid	Uncertain significance
pyrazinamide	Uncertain significance
rifampicin	Uncertain significance
streptomycin	Uncertain significance

GENOMIC VARIANTS RESISTANCE

	Locus	Gene	Size	Position	Genomic Position	Type	WT	Ca
	Rv1918c	PPE35	2964	2687	2167926	Mutation	A	G
	Rv1918c	PPE35	2964	2630	2167983	Mutation	C	T
	Rv0010c	Rv0010c	426	261	13298	Mutation	G	C
	Rv0010c	Rv0010c	426	99	13460	Mutation	A	G
	Rv0565c	Rv0565c	1461	-108	657578	Mutation	A	G
	Rv1129c	Rv1129c	1461	-28	1254562	Mutation	A	G
e	Rv1979c	Rv1979c	1446	857	2222308	Mutation	T	C
e	Rv1979c	Rv1979c	1446	-129	2223293	Mutation	T	C
n,	Rv2477c	Rv2477c	1677	697	2783346	Mutation	G	A
n,	Rv2477c	Rv2477c	1677	396	2783647	Mutation	G	A
	Rv2680	Rv2680	633	-101	2996003	Structural Variant	GTGACCTCCGCCGGCGACGATGCAGAGCGCAGCGATGAGGAGGAGCGGCGCT	G
	Rv2681	Rv2681	1317	174	2996912	Mutation	C	T
	Rv2752c	Rv2752c	1677	1560	3064632	Mutation	G	A
	Rv3083	Rv3083	1488	211	3448714	Mutation	G	C
	Rv3805c	aftB	1884	-410	4269246	Mutation	C	T
	Rv2428	ahpC	588	-142	2726051	Mutation	G	A
	Rv2428	ahpC	588	276	2726468	Mutation	G	A
	Rv2780	ald	1116	-32	3086788	Mutation	T	C
,	Rv1819c	bacA	1920	1807	2062922	Mutation	T	C
	Rv3596c	clpC1	2547	188	4040517	Mutation	A	G
	Rv3794	embA	3285	577	4243809	Mutation	A	G
	Rv3794	embA	3285	616	4243848	Mutation	G	A
	Rv3794	embA	3285	2737	4245969	Mutation	C	T
	Rv3795	embB	3297	823	4247336	Mutation	A	G
	Rv3795	embB	3297	1133	4247646	Mutation	A	C
	Rv3793	embC	3285	809	4240671	Mutation	C	T
	Rv3793	embC	3285	1180	4241042	Mutation	A	G
	Rv3793	embC	3285	2781	4242643	Mutation	C	T

	Locus	Gene	Size	Position	Genomic Position	Type	WT	Ca
	Rv1267c	embR	1167	329	1417019	Mutation	C	T
	Rv1267c	embR	1167	-207	1417554	Mutation	G	C
	Rv1267c	embR	1167	-446	1417793	Mutation	G	A
,	Rv0407	fgd1	1011	960	491742	Mutation	T	C
	Rv3919c	gid	675	615	4407588	Mutation	T	C
	Rv3919c	gid	675	330	4407873	Mutation	C	A
,	Rv3696c	glpK	1554	1379	4138377	Mutation	A	G
	Rv0006	gyrA	2517	61	7362	Mutation	G	C
	Rv0006	gyrA	2517	284	7585	Mutation	G	C
	Rv0006	gyrA	2517	1151	8452	Mutation	C	T
	Rv0006	gyrA	2517	1842	9143	Mutation	T	C
	Rv0006	gyrA	2517	2003	9304	Mutation	G	A
	Rv0005	gyrB	2028	873	6112	Mutation	G	C
	Rv0005	gyrB	2028	885	6124	Mutation	C	T
	Rv2245	kasA	1251	18	2518132	Mutation	C	T
	Rv1908c	katG	2223	1388	2154724	Mutation	C	A
	Rv3244c	lpqB	1752	425	3624486	Mutation	T	C
e	Rv0676c	mmpL5	2895	2842	775639	Mutation	T	C
e	Rv0676c	mmpL5	2895	2381	776100	Mutation	G	A
	Rv3245c	mtrB	1704	1549	3625065	Mutation	T	G
	Rv0639	nusG	717	-138	734116	Mutation	T	C
	Rv0639	nusG	717	324	734577	Mutation	C	T
	Rv2043c	pncA	561	465	2288777	Mutation	C	A
	Rv0701	rplC	654	-452	800357	Mutation	C	A
	Rv3457c	rpoA	1044	-113	3878580	Structural Variant	TGGGGGTACTCGGGGGGCGGCGCCCCCCGAGGGTTGGGTTG	T
	Rv0667	rpoB	3519	3225	763031	Mutation	T	C
	Rv0668	rpoC	3951	515	763884	Mutation	C	T
	Rv0668	rpoC	3951	517	763886	Mutation	C	A
	Rv0668	rpoC	3951	1802	765171	Mutation	C	T
	Rv0682	rpsL	375	-165	781395	Mutation	T	C
,	EBG00000313325	rrs	1537		1471659	Mutation	C	T
id	Rv2754c	thyX	753	455	3067491	Mutation	G	A
	Rv1694	tlyA	807	33	1917972	Mutation	A	G

	Locus	Gene	Size	Position	Genomic Position	Type	WT	Ca
	Rv1644	tsnR	783	369	1853974	Mutation	C	T
	Rv1644	tsnR	783	440	1854045	Mutation	A	G
	Rv1644	tsnR	783	695	1854300	Mutation	T	C
	Rv3806c	ubiA	909	703	4269131	Mutation	G	A
	Rv3806c	ubiA	909	447	4269387	Mutation	T	G
	Rv3806c	ubiA	909	228	4269606	Mutation	A	G
,	Rv3862c	whiB6	351	196	4338326	Mutation	C	T
,	Rv3862c	whiB6	351	-75	4338595	Deletion	GC	G
,	Rv3862c	whiB6	351	-82	4338603	Mutation	G	A
,	Rv3862c	whiB6	351	-211	4338732	Mutation	G	A

MUTATIONS - LINEAGE

Lineage	Locus	Position	WT	Call	Coverage	Relative coverage (%)
lineage1.1.2 (Indo-Oceanic / EAI)	-	12234	-	-	113	100.0
lineage1.1.2 (Indo-Oceanic / EAI)	-	225039	-	-	109	100.0
lineage1.1.2 (Indo-Oceanic / EAI)	-	498010	-	-	122	100.0
lineage1.1.2 (Indo-Oceanic / EAI)	-	559350	-	-	97	100.0
lineage1.1.2 (Indo-Oceanic / EAI)	-	567446	-	-	108	100.0
lineage1.1.2 (Indo-Oceanic / EAI)	-	569045	-	-	96	100.0
lineage1.1.2 (Indo-Oceanic / EAI)	-	711155	-	-	110	100.0
lineage1.1.2 (Indo-Oceanic / EAI)	-	2622402	-	-	122	100.0
lineage1.1.2 (Indo-Oceanic / EAI)	-	3879882	-	-	91	100.0
lineage1.1.2 (Indo-Oceanic / EAI)	-	4157259	-	-	119	100.0

QUALITY METRICS I COVERAGE

Antibiotic	Locus	Gene	QC	Size	Percentage depth > 5	Median Depth	Start	Stop
-	Rv0001	dnaA	PASS	1524	100.0	105.0	1	1524
fluoroquinolones	Rv0005	gyrB	PASS	2028	100.0	128.0	5240	7267
fluoroquinolones	Rv0006	gyrA	PASS	2517	100.0	115.0	7302	9818
-	Rv0010c	Rv0010c	PASS	426	100.0	117.0	13133	13558
delamanid, clofazimine	Rv0407	fgd1	PASS	1011	100.0	117.0	490783	491793
-	Rv0486	mshA	PASS	1443	100.0	134.0	575348	576790
-	Rv0529	ccsA	PASS	975	100.0	141.0	619891	620865
-	Rv0565c	Rv0565c	PASS	1461	100.0	122.0	656010	657470
-	Rv0635	hadA	PASS	477	100.0	107.0	731930	732406
-	Rv0639	nusG	PASS	717	100.0	122.0	734254	734970
-	Rv0643c	mmaA3	PASS	882	100.0	130.0	737268	738149
rifampicin	Rv0667	rpoB	PASS	3519	100.0	109.0	759807	763325
rifampicin	Rv0668	rpoC	PASS	3951	100.0	117.0	763370	767320
bedaquiline, clofazimine	Rv0676c	mmpL5	PASS	2895	100.0	108.0	775586	778480
bedaquiline, clofazimine	Rv0677c	mmpS5	PASS	429	100.0	109.0	778477	778905
bedaquiline, clofazimine	Rv0678	mmpR5	PASS	498	100.0	106.0	778990	779487
streptomycin	Rv0682	rpsL	PASS	375	100.0	109.0	781560	781934
linezolid	Rv0701	rplC	PASS	654	100.0	127.0	800809	801462
-	Rv1129c	Rv1129c	PASS	1461	100.0	133.0	1253074	1254534
delamanid	Rv1173	fbiC	PASS	2571	100.0	111.0	1302931	1305501
multiple	Rv1221	sigE	PASS	774	100.0	115.0	1364413	1365186
-	Rv1258c	Rv1258c	PASS	1260	100.0	123.0	1406081	1407340
ethambutol	Rv1267c	embR	PASS	1167	100.0	124.0	1416181	1417347
bedaquiline	Rv1305	atpE	PASS	246	100.0	121.0	1461045	1461290
aminoglycosides	EBG00000313325	rrs	PASS	1537	100.0	72.0	1471846	1473382
linezolid, capreomycin	EBG00000313339	rrl	PASS	3138	100.0	76.0	1473658	1476795
isoniazid, ethionamide	Rv1484	inhA	PASS	810	100.0	114.0	1674202	1675011
pyrazinamide	Rv1630	rpsA	PASS	1446	100.0	105.0	1833542	1834987
-	Rv1644	tsnR	PASS	783	100.0	133.0	1853606	1854388
capreomycin	Rv1694	tlyA	PASS	807	100.0	117.0	1917940	1918746
-	Rv1819c	bacA	PASS	1920	100.0	110.0	2062809	2064728
isoniazid	Rv1854c	ndh	PASS	1392	100.0	123.0	2101651	2103042
isoniazid	Rv1908c	katG	PASS	2223	100.0	108.0	2153889	2156111
-	Rv1918c	PPE35	PASS	2964	100.0	99.0	2167649	2170612
-	Rv1979c	Rv1979c	PASS	1446	100.0	117.0	2221719	2223164
pyrazinamide	Rv2043c	pncA	PASS	561	100.0	125.0	2288681	2289241
isoniazid	Rv2245	kasA	PASS	1251	100.0	108.0	2518115	2519365
kanamycin, amikacin	Rv2416c	eis	PASS	1209	100.0	123.0	2714124	2715332
isoniazid	Rv2428	ahpC	PASS	588	100.0	113.0	2726193	2726780

Antibiotic	Locus	Gene	QC	Size	Percentage depth > 5	Median Depth	Start	Stop
para-aminosalicylic_acid	Rv2447c	folC	PASS	1464	100.0	124.0	2746135	2747598
-	Rv2477c	Rv2477c	PASS	1677	91.93	107.0	2782366	2784042
bedaquiline, clofazimine	Rv2535c	pepQ	PASS	1119	100.0	123.0	2859300	2860418
para-aminosalicylic_acid	Rv2671	ribD	PASS	777	100.0	118.0	2986839	2987615
-	Rv2680	Rv2680	PASS	633	100.0	125.0	2996105	2996737
-	Rv2681	Rv2681	PASS	1317	100.0	127.0	2996739	2998055
-	Rv2752c	Rv2752c	PASS	1677	100.0	126.0	3064515	3066191
para-aminosalicylic_acid	Rv2754c	thyX	PASS	753	100.0	125.0	3067193	3067945
para-aminosalicylic_acid	Rv2764c	thyA	PASS	792	100.0	122.0	3073680	3074471
-	Rv2780	ald	PASS	1116	100.0	113.0	3086820	3087935
delamanid	Rv2983	fbiD	PASS	645	100.0	130.0	3339118	3339762
-	Rv3083	Rv3083	PASS	1488	100.0	117.0	3448504	3449991
multiple	Rv3197A	whiB7	PASS	279	100.0	115.0	3568401	3568679
-	Rv3236c	Rv3236c	PASS	1158	100.0	126.0	3611959	3613116
-	Rv3244c	lpqB	PASS	1752	100.0	127.0	3623159	3624910
-	Rv3245c	mtrB	PASS	1704	100.0	124.0	3624910	3626613
-	Rv3246c	mtrA	PASS	687	100.0	112.0	3626663	3627349
delamanid	Rv3261	fbiA	PASS	996	100.0	115.0	3640543	3641538
delamanid	Rv3262	fbiB	PASS	1347	100.0	125.0	3641535	3642881
cycloserine	Rv3423c	alr	PASS	1227	100.0	117.0	3840194	3841420
-	Rv3457c	rpoA	PASS	1044	100.0	114.0	3877464	3878507
delamanid	Rv3547	ddn	PASS	456	100.0	121.0	3986844	3987299
pyrazinamide	Rv3596c	clpC1	PASS	2547	100.0	107.0	4038158	4040704
pyrazinamide	Rv3601c	panD	PASS	420	100.0	113.0	4043862	4044281
-	Rv3696c	glpK	PASS	1554	100.0	131.0	4138202	4139755
ethambutol	Rv3793	embC	PASS	3285	100.0	114.0	4239863	4243147
ethambutol	Rv3794	embA	PASS	3285	100.0	137.0	4243233	4246517
ethambutol	Rv3795	embB	PASS	3297	100.0	120.0	4246514	4249810
-	Rv3805c	aftB	PASS	1884	100.0	129.0	4266953	4268836
ethambutol	Rv3806c	ubiA	PASS	909	100.0	146.0	4268925	4269833
ethionamide	Rv3854c	ethA	PASS	1470	100.0	129.0	4326004	4327473
ethionamide	Rv3855	ethR	PASS	651	100.0	127.0	4327549	4328199
multiple	Rv3862c	whiB6	PASS	351	100.0	116.0	4338171	4338521
streptomycin	Rv3919c	gid	PASS	675	100.0	121.0	4407528	4408202

QUALITY METRICS II RESISTANCE

Antibiotic	Locus	Gene	QC	NA mutations	AA mutations
-	Rv0001	dnaA	PASS	0	0
fluoroquinolones	Rv0005	gyrB	PASS	2	2
fluoroquinolones	Rv0006	gyrA	PASS	5	5
-	Rv0010c	Rv0010c	PASS	2	2
delamanid, clofazimine	Rv0407	fgd1	PASS	1	1
-	Rv0486	mshA	PASS	0	0
-	Rv0529	ccsA	PASS	0	0
-	Rv0565c	Rv0565c	PASS	1	0
-	Rv0635	hadA	PASS	0	0
-	Rv0639	nusG	PASS	2	1
-	Rv0643c	mmaA3	PASS	0	0
rifampicin	Rv0667	rpoB	PASS	1	1
rifampicin	Rv0668	rpoC	PASS	3	3
bedaquiline, clofazimine	Rv0676c	mmpL5	PASS	2	2
bedaquiline, clofazimine	Rv0677c	mmpS5	PASS	0	0
bedaquiline, clofazimine	Rv0678	mmpR5	PASS	0	0
streptomycin	Rv0682	rpsL	PASS	1	0
linezolid	Rv0701	rplC	PASS	1	0
-	Rv1129c	Rv1129c	PASS	1	0
delamanid	Rv1173	fbiC	PASS	0	0
multiple	Rv1221	sigE	PASS	0	0
-	Rv1258c	Rv1258c	PASS	0	0
ethambutol	Rv1267c	embR	PASS	3	1
bedaquiline	Rv1305	atpE	PASS	0	0
aminoglycosides	EBG00000313325	rrs	PASS	1	0
linezolid, capreomycin	EBG00000313339	rrl	PASS	0	0
isoniazid, ethionamide	Rv1484	inhA	PASS	0	0
pyrazinamide	Rv1630	rpsA	PASS	0	0
-	Rv1644	tsnR	PASS	3	3
capreomycin	Rv1694	tlyA	PASS	1	1
-	Rv1819c	bacA	PASS	1	1
isoniazid	Rv1854c	ndh	PASS	0	0
isoniazid	Rv1908c	katG	PASS	1	1
-	Rv1918c	PPE35	PASS	2	2
-	Rv1979c	Rv1979c	PASS	2	1
pyrazinamide	Rv2043c	pncA	PASS	1	1
isoniazid	Rv2245	kasA	PASS	1	1
kanamycin, amikacin	Rv2416c	eis	PASS	0	0
isoniazid	Rv2428	ahpC	PASS	2	1
para-aminosalicylic_acid	Rv2447c	folC	PASS	0	0

Antibiotic	Locus	Gene	QC	NA mutations	AA mutations
-	Rv2477c	Rv2477c	PASS	2	2
bedaquiline, clofazimine	Rv2535c	pepQ	PASS	0	0
para-aminosalicylic_acid	Rv2671	ribD	PASS	0	0
-	Rv2680	Rv2680	PASS	1	0
-	Rv2681	Rv2681	PASS	1	1
-	Rv2752c	Rv2752c	PASS	1	1
para-aminosalicylic_acid	Rv2754c	thyX	PASS	1	1
para-aminosalicylic_acid	Rv2764c	thyA	PASS	0	0
-	Rv2780	ald	PASS	1	0
delamanid	Rv2983	fbiD	PASS	0	0
-	Rv3083	Rv3083	PASS	1	1
multiple	Rv3197A	whiB7	PASS	0	0
-	Rv3236c	Rv3236c	PASS	0	0
-	Rv3244c	lpqB	PASS	1	1
-	Rv3245c	mtrB	PASS	1	1
-	Rv3246c	mtrA	PASS	0	0
delamanid	Rv3261	fbiA	PASS	0	0
delamanid	Rv3262	fbiB	PASS	0	0
cycloserine	Rv3423c	alr	PASS	0	0
-	Rv3457c	rpoA	PASS	1	0
delamanid	Rv3547	ddn	PASS	0	0
pyrazinamide	Rv3596c	clpC1	PASS	1	1
pyrazinamide	Rv3601c	panD	PASS	0	0
-	Rv3696c	glpK	PASS	1	1
ethambutol	Rv3793	embC	PASS	3	3
ethambutol	Rv3794	embA	PASS	3	3
ethambutol	Rv3795	embB	PASS	2	2
-	Rv3805c	aftB	PASS	1	0
ethambutol	Rv3806c	ubiA	PASS	3	3
ethionamide	Rv3854c	ethA	PASS	0	0
ethionamide	Rv3855	ethR	PASS	0	0
multiple	Rv3862c	whiB6	PASS	4	1
streptomycin	Rv3919c	gid	PASS	2	2

QUALITY METRICS III RESISTANCE

Antibiotic	Locus	Gene	QC	NA mutations	AA mutations	Unknown mutations	Number of deletions	Unknown deletions	Number of insertions	Unknown insertions
-	Rv0001	dnaA	PASS	0	0	0	0	0	0	0
fluoroquinolones	Rv0005	gyrB	PASS	2	2	2	0	0	0	0
fluoroquinolones	Rv0006	gyrA	PASS	5	5	5	0	0	0	0
-	Rv0010c	Rv0010c	PASS	2	2	2	0	0	0	0
delamanid, clofazimine	Rv0407	fgd1	PASS	1	1	1	0	0	0	0
-	Rv0486	mshA	PASS	0	0	0	0	0	0	0
-	Rv0529	ccsA	PASS	0	0	0	0	0	0	0
-	Rv0565c	Rv0565c	PASS	1	0	1	0	0	0	0
-	Rv0635	hadA	PASS	0	0	0	0	0	0	0
-	Rv0639	nusG	PASS	2	1	2	0	0	0	0
-	Rv0643c	mmaA3	PASS	0	0	0	0	0	0	0
rifampicin	Rv0667	rpoB	PASS	1	1	1	0	0	0	0
rifampicin	Rv0668	rpoC	PASS	3	3	3	0	0	0	0
bedaquiline, clofazimine	Rv0676c	mmpL5	PASS	2	2	2	0	0	0	0
bedaquiline, clofazimine	Rv0677c	mmpS5	PASS	0	0	0	0	0	0	0
bedaquiline, clofazimine	Rv0678	mmpR5	PASS	0	0	0	0	0	0	0
streptomycin	Rv0682	rpsL	PASS	1	0	1	0	0	0	0
linezolid	Rv0701	rplC	PASS	1	0	1	0	0	0	0
-	Rv1129c	Rv1129c	PASS	1	0	1	0	0	0	0
delamanid	Rv1173	fbiC	PASS	0	0	0	0	0	0	0
multiple	Rv1221	sigE	PASS	0	0	0	0	0	0	0
-	Rv1258c	Rv1258c	PASS	0	0	0	0	0	0	0
ethambutol	Rv1267c	embR	PASS	3	1	3	0	0	0	0
bedaquiline	Rv1305	atpE	PASS	0	0	0	0	0	0	0
aminoglycosides	EBG00000313325	rrs	PASS	1	0	1	0	0	0	0
linezolid, capreomycin	EBG00000313339	rrl	PASS	0	0	0	0	0	0	0
isoniazid, ethionamide	Rv1484	inhA	PASS	0	0	0	0	0	0	0
pyrazinamide	Rv1630	rpsA	PASS	0	0	0	0	0	0	0
-	Rv1644	tsnR	PASS	3	3	3	0	0	0	0
capreomycin	Rv1694	tlyA	PASS	1	1	1	0	0	0	0
-	Rv1819c	bacA	PASS	1	1	1	0	0	0	0
isoniazid	Rv1854c	ndh	PASS	0	0	0	0	0	0	0
isoniazid	Rv1908c	katG	PASS	1	1	1	0	0	0	0
-	Rv1918c	PPE35	PASS	2	2	2	0	0	0	0
-	Rv1979c	Rv1979c	PASS	2	1	2	0	0	0	0

Antibiotic	Locus	Gene	QC	NA mutations	AA mutations	Unknown mutations	Number of deletions	Unknown deletions	Number of insertions	Unknown insertions
pyrazinamide	Rv2043c	pncA	PASS	1	1	1	0	0	0	0
isoniazid	Rv2245	kasA	PASS	1	1	1	0	0	0	0
kanamycin, amikacin	Rv2416c	eis	PASS	0	0	0	0	0	0	0
isoniazid	Rv2428	ahpC	PASS	2	1	2	0	0	0	0
para-aminosalicylic_acid	Rv2447c	folC	PASS	0	0	0	0	0	0	0
-	Rv2477c	Rv2477c	PASS	2	2	2	0	0	0	0
bedaquiline, clofazimine	Rv2535c	pepQ	PASS	0	0	0	0	0	0	0
para-aminosalicylic_acid	Rv2671	ribD	PASS	0	0	0	0	0	0	0
-	Rv2680	Rv2680	PASS	1	0	1	1	1	0	0
-	Rv2681	Rv2681	PASS	1	1	1	0	0	0	0
-	Rv2752c	Rv2752c	PASS	1	1	1	0	0	0	0
para-aminosalicylic_acid	Rv2754c	thyX	PASS	1	1	1	0	0	0	0
para-aminosalicylic_acid	Rv2764c	thyA	PASS	0	0	0	0	0	0	0
-	Rv2780	ald	PASS	1	0	1	0	0	0	0
delamanid	Rv2983	fbiD	PASS	0	0	0	0	0	0	0
-	Rv3083	Rv3083	PASS	1	1	1	0	0	0	0
multiple	Rv3197A	whiB7	PASS	0	0	0	0	0	0	0
-	Rv3236c	Rv3236c	PASS	0	0	0	0	0	0	0
-	Rv3244c	lpqB	PASS	1	1	1	0	0	0	0
-	Rv3245c	mtrB	PASS	1	1	1	0	0	0	0
-	Rv3246c	mtrA	PASS	0	0	0	0	0	0	0
delamanid	Rv3261	fbiA	PASS	0	0	0	0	0	0	0
delamanid	Rv3262	fbiB	PASS	0	0	0	0	0	0	0
cycloserine	Rv3423c	alr	PASS	0	0	0	0	0	0	0
-	Rv3457c	rpoA	PASS	1	0	1	1	1	0	0
delamanid	Rv3547	ddn	PASS	0	0	0	0	0	0	0
pyrazinamide	Rv3596c	clpC1	PASS	1	1	1	0	0	0	0
pyrazinamide	Rv3601c	panD	PASS	0	0	0	0	0	0	0
-	Rv3696c	glpK	PASS	1	1	1	0	0	0	0
ethambutol	Rv3793	embC	PASS	3	3	3	0	0	0	0
ethambutol	Rv3794	embA	PASS	3	3	3	0	0	0	0
ethambutol	Rv3795	embB	PASS	2	2	2	0	0	0	0
-	Rv3805c	aftB	PASS	1	0	1	0	0	0	0
ethambutol	Rv3806c	ubiA	PASS	3	3	3	0	0	0	0
ethionamide	Rv3854c	ethA	PASS	0	0	0	0	0	0	0
ethionamide	Rv3855	ethR	PASS	0	0	0	0	0	0	0
multiple	Rv3862c	whiB6	PASS	4	1	4	1	1	0	0
streptomycin	Rv3919c	gid	PASS	2	2	2	0	0	0	0

SIGNIFICANT VIRULENCE FACTORS

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_103772	2856	3689	+	lipF	1-834/834	0/0	100.0	100.0	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307
k141_121393	9379	10083	+	mgtC	1-705/705	0/0	100.0	100.0	NP_216327	(mgtC) Possible Mg2+ transport P-type ATPase C MgtC [MgtC (VF0289)] [Mycobacterium tuberculosis H37Rv]	VF0289
k141_144217	795	1697	+	Rv1794	1-903/903	0/0	100.0	100.0	NP_216310	(Rv1794) ESX-5 locus protein [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_144217	1969	3480	+	eccD5	1-1512/1512	0/0	100.0	100.0	NP_216311	(eccD5) ESX-5 type VII secretion system EccD5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_16303	1754	2047	-	esxG	1-294/294	0/0	100.0	100.0	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_230642	4654	5631	-	fbpB	1-978/978	0/0	100.0	100.0	NP_216402	(fbpB) Secreted antigen 85-B FbpB (85B) (antigen 85 complex B) (mycolyl transferase 85B) (fibronectin-binding protein B) (extracellular alpha-antigen) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_250635	236	1933	+	mbtA	1-1698/1698	0/0	100.0	100.0	NP_216900	(mbtA) Bifunctional enzyme MbtA: salicyl-AMP ligase (SAL-AMP ligase) + salicyl-S-ArCP synthetase [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_250635	2028	2948	+	mbtJ	1-921/921	0/0	100.0	100.0	YP_177876	(mbtJ) Putative acetyl hydrolase MbtJ [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_34847	10680	11966	+	icl	1-1287/1287	0/0	100.0	100.0	YP_177728	(icl) Isocitrate lyase Icl (isocitrase) (isocitratase) [Isocitrate lyase (VF0253)] [Mycobacterium tuberculosis H37Rv]	VF0253
k141_65643	4898	6058	+	mbtN	1-1161/1161	0/0	100.0	100.0	NP_215862	(mbtN) Acyl-CoA dehydrogenase MbtN [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_65643	6025	6657	-	mbtK	1-633/633	0/0	100.0	100.0	NP_215863	(mbtK) Lysine N-acetyltransferase MbtK [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_70144	1373	1660	+	esxA	1-288/288	0/0	100.0	100.0	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_96128	1906	2217	-	espC	1-312/312	0/0	100.0	100.0	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_65643	3333	4898	+	mbtM	1-1566/1566	0/0	100.0	99.94	NP_215861	(mbtM) Probable fatty acyl-AMP ligase MbtM [FadD33 (VF0308)] [Mycobacterium tuberculosis H37Rv]	VF0308
k141_186627	8632	10089	-	phoR	1-1458/1458	0/0	100.0	99.93	NP_215272	(phoR) Possible two component system response sensor kinase membrane associated PhoR [PhoP/R (CVF331)] [Mycobacterium tuberculosis H37Rv]	NA
k141_16303	2096	3638	-	PPE4	1-1542/1542	1/1	100.0	99.87	YP_177711	(PPE4) PPE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_70144	3721	5256	+	eccD1	1-1536/1536	0/0	100.0	99.87	NP_218394	(eccD1) ESX-1 type VII secretion system EccD1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_122201	408	1100	+	ideR	1-693/693	0/0	100.0	99.86	NP_217227	(ideR) Iron-dependent repressor and activator IdeR [IdeR (VF0300)] [Mycobacterium tuberculosis H37Rv]	VF0300
k141_3693	2073	2657	-	PPE41	1-585/585	0/0	100.0	99.83	YP_177881	(PPE41) PPE family protein PPE41 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_16303	536	1423	-	espG3	1-888/888	0/0	100.0	99.78	NP_214803	(espG3) Type VII secretion system protein EspG3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_65033	3543	3842	+	PE35	1-300/300	0/0	100.0	99.67	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_70144	1038	1340	+	esxB	1-303/303	0/0	100.0	99.67	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_16303	1434	1724	-	esxH	1-291/291	0/0	100.0	99.66	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_250247	2054	2269	+	mbtH	1-216/216	0/0	100.0	99.54	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_7807	98	533	-	hspX	1-435/435	1/1	100.0	99.54	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301
k141_65643	3020	3340	+	mbtL	1-321/321	0/0	100.0	99.38	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_16303	3641	3949	-	PE5	1-309/309	0/0	100.0	99.03	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_58753	44	344	+	PE35	1-300/300	1/1	100.0	98.34	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_90344	13	303	-	esxA	1-288/288	1/3	100.0	98.28	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_172091	167	467	+	PE35	1-300/300	1/1	100.0	97.67	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_11656	493	708	+	mbtH	1-216/216	0/0	100.0	97.22	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_144217	423	707	+	esxN	1-285/285	0/0	100.0	93.68	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_49638	330	615	-	esxN	1-285/285	1/1	100.0	90.91	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_98178	188	475	+	esxN	1-285/285	1/3	100.0	90.28	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_164440	559	851	-	esxH	1-291/291	1/2	100.0	83.96	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_248709	4522	6354	-	eccA5	1-1833/1833	2/2	99.95	99.84	NP_216314	(eccA5) ESX-5 type VII secretion system AAA+ ATPase EccA5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_65643	9672	11410	+	irtB	1-1740/1740	1/1	99.94	99.94	NP_215865	(irtB) Iron-regulated transporter IrtB [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_17232	4651	5984	-	mbtC	1-1335/1335	1/1	99.93	99.92	NP_216898	(mbtC) Polyketide synthetase MbtC (polyketide synthase) [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_17232	1639	4651	-	mbtD	1-3015/3015	2/2	99.93	99.83	NP_216897	(mbtD) Polyketide synthetase MbtD (polyketide synthase) [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_15064	2586	4110	+	eccB5	1-1521/1521	3/6	99.93	99.41	NP_216298	(eccB5) ESX-5 type VII secretion system protein EccB5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_39407	3545	5915	-	relA	1-2373/2373	1/2	99.92	99.75	NP_217099	(relA) Probable GTP pyrophosphokinase RelA (ATP:GTP 3'-pyrophosphotransferase) (PPGPP synthetase I) ((P)PPGPP synthetase) (GTP diphosphokinase) [RelA (VF0287)] [Mycobacterium tuberculosis H37Rv]	VF0287
k141_70144	6311	8540	-	espK	1-2190/2190	4/44	99.91	97.81	NP_218396	(espK) ESX-1 type VII secretion system EspK [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_113088	1440	2463	+	fbpC	1-1023/1023	2/3	99.9	99.71	YP_177694	(fbpC) Secreted antigen 85-C FbpC (85C) (antigen 85 complex C) (AG58C) (mycolyl transferase 85C) (fibronectin-binding protein C) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_144217	3458	5214	+	mycP5	1-1758/1758	3/3	99.89	99.55	NP_216312	(mycP5) ESX-5 type VII secretion system subtilisin-like serine protease MycP5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_65643	7098	9675	+	irtA	1-2580/2580	2/4	99.88	99.77	NP_215864	(irtA) Iron-regulated transporter IrtA [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_117049	6199	7053	+	erp	1-855/855	2/2	99.88	99.65	NP_218327	(erp) exported repetitive protein precursor Erp [Erp (VF0312)] [Mycobacterium tuberculosis H37Rv]	VF0312
k141_186627	10134	10876	-	phoP	1-744/744	1/1	99.87	99.87	NP_215271	(phoP) Possible two component system response transcriptional positive regulator PhoP [PhoP (VF0286)] [Mycobacterium tuberculosis H37Rv]	VF0286
k141_15064	5483	8277	+	eccCb5	1-2799/2799	4/4	99.86	99.71	NP_216300	(eccCb5) ESX-5 type VII secretion system FtsK/SpoIIIE family protein EccCb5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_250635	2952	4302	-	mbtI	1-1353/1353	1/2	99.85	99.7	YP_177877	(mbtI) Isochorismate synthase MbtI [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_188587	3279	5171	+	eccA3	1-1896/1896	1/3	99.84	99.74	NP_214796	(eccA3) Type VII secretion system protein EccA3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_15064	4107	5411	+	eccCa5	1-1306/1308	1/1	99.77	99.92	NP_216299	(eccCa5) ESX-5 type VII secretion system FtsK/SpoIIIE family protein EccCa5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_250247	783	2076	+	mbtG	1-1296/1296	3/4	99.77	99.46	NP_216894	(mbtG) Lysine-N-oxygenase MbtG (L-lysine 6-monooxygenase) (lysine N6-hydroxylase) [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_65033	1628	3400	+	eccCb1	1-1776/1776	4/5	99.77	99.33	NP_218388	(eccCb1) ESX-1 type VII secretion system FtsK/SpoIIIE family protein EccCb1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_242688	313	1054	+	phoP	1-744/744	2/2	99.73	98.66	NP_215271	(phoP) Possible two component system response transcriptional positive regulator PhoP [PhoP (VF0286)] [Mycobacterium tuberculosis H37Rv]	VF0286

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_189330	1020	2033	+	fbpA	1-1017/1017	3/3	99.71	98.53	NP_218321	(fbpA) Secreted antigen 85-a FbpA (mycolyl transferase 85A) (fibronectin-binding protein A) (antigen 85 complex A) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_151637	1053	1370	+	espC	1-312/312	5/8	99.68	93.73	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_49614	464	1065	-	hbhA	1-600/600	6/6	99.67	96.69	NP_214989	(hbhA) iron-regulated heparin binding hemagglutinin hbhA (adhesin) [HbhA (VF0313)] [Mycobacterium tuberculosis H37Rv]	VF0313
k141_221656	4	305	-	PE35	1-300/300	2/4	99.67	96.04	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_219998	176	476	+	PE35	1-300/300	2/3	99.67	96.03	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_122578	646	941	-	esxM	1-297/297	1/1	99.66	97.31	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_149970	143	436	-	esxG	1-294/294	2/2	99.66	97.29	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_99397	799	1096	+	esxG	1-294/294	3/6	99.66	96.66	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_41534	170	466	-	esxG	1-294/294	4/5	99.66	95.97	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_194257	108	403	-	esxG	1-294/294	3/4	99.66	95.96	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_91237	54	350	+	esxM	1-296/297	1/1	99.66	95.96	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_253649	236	532	+	esxH	1-291/291	4/8	99.66	95.64	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_210068	268	564	-	esxM	1-296/297	1/1	99.66	94.61	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_54650	248	550	+	esxM	1-296/297	3/7	99.66	93.73	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_199341	677	972	-	esxM	2-297/297	0/0	99.66	83.45	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_137680	715	1008	-	esxH	1-291/291	3/5	99.66	83.05	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_40186	510	801	-	esxH	1-291/291	2/3	99.66	82.25	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_54527	837	2552	-	eccA1	1-1722/1722	2/6	99.65	99.36	NP_218385	(eccA1) ESX-1 type VII secretion system AAA+ ATPase EccA1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_163056	414	697	-	esxN	1-285/285	1/1	99.65	97.19	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_27842	378	661	-	esxN	1-285/285	1/1	99.65	92.28	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_199341	379	662	-	esxN	1-284/285	0/0	99.65	83.1	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_51383	1194	2531	-	mycP1	1-1337/1341	2/3	99.63	99.7	NP_218400	(mycP1) ESX-1 type VII secretion system subtilisin-like serine protease MycP1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_92076	779	1772	+	eccE3	1-996/996	5/6	99.6	97.59	NP_214806	(eccE3) Type VII secretion system transmembrane protein EccE3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_231351	15	234	-	mbtH	1-216/216	3/6	99.54	96.38	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_111524	300	1282	-	fbpB	1-978/978	9/15	99.49	96.76	NP_216402	(fbpB) Secreted antigen 85-B FbpB (85B) (antigen 85 complex B) (mycolyl transferase 85B) (fibronectin-binding protein B) (extracellular alpha-antigen) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_96128	1239	1790	-	espD	1-555/555	2/3	99.46	99.1	NP_218131	(espD) ESX-1 type VII secretion system EspD [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_179306	37	358	-	mbtL	1-321/321	4/5	99.38	94.75	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_144217	98	394	+	esxM	1-295/297	2/2	99.33	96.3	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_133100	240	534	+	esxM	1-297/297	1/2	99.33	94.95	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_25444	479	789	-	PE35	1-300/300	8/15	99.33	94.25	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_85995	311	913	-	hbhA	1-600/600	8/11	99.33	94.07	NP_214989	(hbhA) iron-regulated heparin binding hemagglutinin hbhA (adhesin) [HbhA (VF0313)] [Mycobacterium tuberculosis H37Rv]	VF0313
k141_55980	76	372	+	esxM	2-297/297	3/3	99.33	80.54	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_79981	199	493	+	esxG	1-294/294	3/5	99.32	94.61	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_242819	692	978	-	esxA	1-288/288	3/3	99.31	97.58	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_184589	366	652	+	esxA	1-288/288	2/3	99.31	96.89	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_242400	45	334	+	esxA	1-288/288	4/6	99.31	93.84	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_168704	316	607	+	esxA	1-288/288	5/8	99.31	93.54	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_253735	699	989	-	esxH	1-291/291	3/4	99.31	84.3	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_133100	583	866	+	esxN	1-285/285	3/3	99.3	97.2	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_234771	1271	1555	+	esxN	1-283/285	2/2	99.3	93.68	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_128986	212	497	+	esxN	1-283/285	1/3	99.3	93.36	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_140260	43	326	+	esxN	1-283/285	1/1	99.3	91.9	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_39195	280	563	-	esxN	1-285/285	3/3	99.3	91.26	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_257139	117	401	-	esxN	1-283/285	2/2	99.3	91.23	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_182229	155	443	-	esxN	1-283/285	1/6	99.3	91.0	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_64582	341	627	-	esxN	1-283/285	3/4	99.3	90.94	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_257839	562	846	-	esxN	1-285/285	2/4	99.3	87.81	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_233	1089	1784	-	ideR	1-693/693	6/13	99.28	96.01	NP_217227	(ideR) Iron-dependent repressor and activator IdeR [IdeR (VF0300)] [Mycobacterium tuberculosis H37Rv]	VF0300
k141_87767	55	645	+	PPE41	1-585/585	9/16	99.15	95.97	YP_177881	(PPE41) PPE family protein PPE41 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_158928	525	846	+	mbtL	1-321/321	5/7	99.07	96.61	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_195797	203	512	-	espC	1-312/312	3/4	99.04	97.76	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_224265	92	398	+	PE5	1-309/309	2/4	99.03	97.1	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_145381	778	1088	+	PE5	2-309/309	4/7	99.03	95.85	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_245523	324	625	-	esxB	4-303/303	1/2	99.01	97.02	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_239721	824	1119	+	esxM	1-297/297	2/5	98.99	95.65	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_47972	50	344	+	esxM	1-295/297	2/2	98.99	95.61	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_186602	1024	1319	-	esxM	1-296/297	3/4	98.99	94.3	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_246767	2116	2411	-	esxM	1-296/297	4/4	98.99	93.96	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_118516	768	1468	+	ideR	1-693/693	11/22	98.99	93.36	NP_217227	(ideR) Iron-dependent repressor and activator IdeR [IdeR (VF0300)] [Mycobacterium tuberculosis H37Rv]	VF0300

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_54160	494	793	-	esxM	1-295/297	4/7	98.99	93.36	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_92305	57	355	+	esxM	1-297/297	4/8	98.99	93.05	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_158145	313	606	+	esxM	1-296/297	1/2	98.99	92.91	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_45034	470	764	-	esxM	2-297/297	3/3	98.99	82.16	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_190809	209	499	-	esxG	1-294/294	2/3	98.98	97.62	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_88042	518	809	+	esxG	1-294/294	2/4	98.98	96.27	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_168020	313	601	-	esxH	3-290/291	1/1	98.97	98.62	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_188547	400	688	+	esxH	1-291/291	4/4	98.97	83.9	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_164543	766	1055	-	esxH	1-291/291	3/5	98.97	81.57	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_250312	882	1171	-	esxH	1-291/291	2/5	98.97	80.89	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_130056	876	2307	+	eccB1	1-1439/1443	13/15	98.96	97.64	NP_218386	(eccB1) ESX-1 type VII secretion system protein EccB1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_131221	2021	2302	+	esxN	1-283/285	1/1	98.95	97.53	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_168461	36	317	-	esxN	1-283/285	1/1	98.95	95.41	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_62282	111	393	+	esxN	1-285/285	2/4	98.95	95.11	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_133790	260	541	+	esxN	1-283/285	1/1	98.95	93.64	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_184367	51	332	-	esxN	1-283/285	1/1	98.95	92.23	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_176012	409	691	+	esxN	1-285/285	4/4	98.95	90.56	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_159646	6	287	+	esxN	1-285/285	1/3	98.95	90.17	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_253877	612	894	-	esxN	1-282/285	1/1	98.95	89.75	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_78043	154	437	+	esxN	1-284/285	4/4	98.95	81.82	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_239135	120	402	+	esxN	1-284/285	3/3	98.95	80.35	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_209843	1379	3127	+	mycP5	1-1758/1758	20/29	98.92	96.55	NP_216312	(mycP5) ESX-5 type VII secretion system subtilisin-like serine protease MycP5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_63738	145	693	+	espD	1-555/555	4/6	98.92	95.68	NP_218131	(espD) ESX-1 type VII secretion system EspD [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_153124	412	1332	+	mbtJ	1-921/921	12/20	98.91	95.6	YP_177876	(mbtJ) Putative acetyl hydrolase MbtJ [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_29635	2236	2834	+	hbhA	1-600/600	10/13	98.83	95.05	NP_214989	(hbhA) iron-regulated heparin binding hemagglutinin hbhA (adhesin) [HbhA (VF0313)] [Mycobacterium tuberculosis H37Rv]	VF0313
k141_155778	351	1654	-	eccCa5	1-1306/1308	18/26	98.78	94.84	NP_216299	(eccCa5) ESX-5 type VII secretion system FtsK/SpoIIIE family protein EccCa5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_193826	18	566	+	espD	1-555/555	5/8	98.74	97.3	NP_218131	(espD) ESX-1 type VII secretion system EspD [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_133023	242	549	-	espC	1-312/312	3/4	98.72	98.72	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_102315	67	1789	+	eccA1	1-1722/1722	30/45	98.72	94.44	NP_218385	(eccA1) ESX-1 type VII secretion system AAA+ ATPase EccA1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_184589	34	333	+	esxB	1-303/303	3/5	98.68	96.71	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_71196	30	329	-	esxB	1-303/303	3/5	98.68	96.38	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_26748	1788	2089	+	esxB	1-303/303	3/7	98.68	95.1	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_113925	1	293	-	esxM	1-293/297	0/0	98.65	96.93	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_165956	1130	1423	+	esxM	1-296/297	3/4	98.65	94.61	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_83248	775	1069	-	esxM	1-297/297	4/6	98.65	94.31	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_239998	168	467	-	esxM	1-296/297	4/10	98.65	93.73	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_131150	2093	2385	-	esxM	1-295/297	1/2	98.65	90.85	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_172894	898	1193	-	esxM	2-297/297	4/6	98.65	81.94	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_155398	53	1592	-	PPE4	1-1542/1542	27/40	98.64	96.28	YP_177711	(PPE4) PPE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_155334	485	775	-	esxG	1-294/294	4/5	98.64	95.25	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_164440	888	1178	-	esxG	1-290/294	1/1	98.64	83.85	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_105661	1737	2023	+	esxH	1-291/291	3/4	98.63	82.82	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_237392	748	1039	+	esxH	1-291/291	7/9	98.63	82.09	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_135120	795	1007	-	mbtH	1-213/216	0/0	98.61	97.65	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_33905	357	570	-	mbtH	1-216/216	2/4	98.61	97.23	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_252191	508	796	+	esxA	1-288/288	3/9	98.61	96.25	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_151209	257	537	-	esxN	1-283/285	1/2	98.6	93.29	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_53394	2	284	+	esxN	1-285/285	3/6	98.6	91.64	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_107539	159	439	+	esxN	1-282/285	1/1	98.6	90.78	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_19385	125	406	+	esxN	1-285/285	3/5	98.6	90.21	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_27830	374	656	-	esxN	1-285/285	4/6	98.6	89.89	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_41457	1487	2189	+	mgtC	1-705/705	12/18	98.58	93.83	NP_216327	(mgtC) Possible Mg2+ transport P-type ATPase C MgtC [MgtC (VF0289)] [Mycobacterium tuberculosis H37Rv]	VF0289
k141_142554	85	400	-	mbtL	1-321/321	2/5	98.44	96.57	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_96798	113	432	-	mbtL	1-321/321	3/9	98.44	94.77	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_105169	102	417	+	mbtL	1-321/321	5/5	98.44	94.7	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_191879	524	1397	-	espG3	1-888/888	9/14	98.42	95.83	NP_214803	(espG3) Type VII secretion system protein EspG3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_248507	97	724	-	mbtK	3-633/633	7/13	98.42	94.97	NP_215863	(mbtK) Lysine N-acetyltransferase MbtK [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_142740	721	1028	+	espC	1-312/312	5/6	98.4	95.53	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_190178	1279	1592	+	espC	1-309/312	6/9	98.4	94.3	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_82747	856	1286	+	hspX	1-435/435	6/10	98.39	95.2	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301
k141_201037	1474	2021	-	espD	1-555/555	8/11	98.38	94.43	NP_218131	(espD) ESX-1 type VII secretion system EspD [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_252191	173	473	+	esxB	1-300/303	4/5	98.35	97.03	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_211924	234	536	-	esxB	1-303/303	6/10	98.35	95.78	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_182287	372	677	+	esxB	1-303/303	9/13	98.35	94.21	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_90337	332	634	-	esxB	1-303/303	7/10	98.35	94.16	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_141881	37	1411	+	mycP3	1-1386/1386	23/35	98.34	94.13	NP_214805	(mycP3) Type VII secretion system membrane-anchored mycosin MycP3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_36789	276	567	-	esxM	1-293/297	1/1	98.32	97.61	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_146068	1	292	-	esxM	1-293/297	1/1	98.32	97.27	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_86558	80	373	-	esxM	1-296/297	4/6	98.32	92.28	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_73600	208	508	-	esxM	1-297/297	7/14	98.32	91.83	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_99094	3988	4279	-	esxM	1-297/297	4/5	98.32	90.24	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_220320	160	464	-	esxM	2-295/297	9/15	98.32	84.69	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_53344	79	374	-	esxM	2-296/297	6/7	98.32	81.61	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_204744	57	348	-	esxM	2-297/297	4/4	98.32	81.08	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_29433	803	1097	+	esxG	1-294/294	5/11	98.3	91.33	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_237392	423	712	+	esxG	1-290/294	2/2	98.3	85.57	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_17203	867	1449	+	PPE41	1-585/585	11/18	98.29	92.75	YP_177881	(PPE41) PPE family protein PPE41 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_256558	45	327	+	esxA	1-288/288	3/5	98.26	97.22	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_245523	8	291	-	esxA	1-288/288	3/6	98.26	96.54	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_55049	26	317	-	esxA	3-288/288	6/12	98.26	93.22	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_105675	586	865	-	esxN	1-285/285	2/5	98.25	94.39	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_4832	197	477	+	esxN	1-283/285	3/4	98.25	92.96	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_179684	33	312	-	esxN	1-285/285	4/5	98.25	90.53	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_234872	478	759	-	esxN	1-285/285	6/7	98.25	89.2	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_100352	250	530	+	esxN	1-285/285	3/6	98.25	88.46	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_174387	362	646	-	esxN	1-281/285	3/6	98.25	88.46	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_61677	170	453	-	esxN	1-285/285	5/9	98.25	84.78	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_160851	173	452	-	esxN	1-284/285	2/4	98.25	80.63	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_169945	663	1971	-	eccCa5	1-1306/1308	28/45	98.24	91.81	NP_216299	(eccCa5) ESX-5 type VII secretion system FtsK/SpoIIIE family protein EccCa5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_28535	647	1546	+	Rv1794	1-903/903	20/29	98.23	93.56	NP_216310	(Rv1794) ESX-5 locus protein [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_27037	1204	2371	+	espA	1-1179/1179	19/31	98.22	93.94	NP_218133	(espA) ESX-1 type VII secretion system EspA [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_145876	13	1002	+	eccE3	1-996/996	18/30	98.19	95.14	NP_214806	(eccE3) Type VII secretion system transmembrane protein EccE3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_253278	37	471	+	hspX	1-435/435	9/16	98.16	93.91	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301
k141_76984	455	891	-	hspX	1-435/435	8/18	98.16	92.58	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301
k141_194724	69	280	+	mbtH	1-216/216	3/4	98.15	97.22	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_105588	169	383	-	mbtH	1-216/216	3/7	98.15	95.43	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_157989	117	431	+	mbtL	4-321/321	2/3	98.13	96.86	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_90021	594	908	+	mbtL	1-321/321	6/6	98.13	96.26	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_208125	457	774	+	mbtL	1-321/321	6/9	98.13	94.75	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_122453	21	1391	+	mycP3	1-1386/1386	25/37	98.12	94.78	NP_214805	(mycP3) Type VII secretion system membrane-anchored mycosin MycP3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_164087	232	781	+	espD	1-555/555	8/17	98.02	95.54	NP_218131	(espD) ESX-1 type VII secretion system EspD [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_137305	40	333	-	PE35	1-300/300	4/6	98.0	96.33	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_114469	269	562	-	esxM	1-295/297	6/7	97.98	94.63	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_206764	679	969	+	esxM	1-296/297	4/5	97.98	90.2	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_15017	528	1411	-	espG3	1-888/888	17/32	97.97	94.12	NP_214803	(espG3) Type VII secretion system protein EspG3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_241546	671	959	+	esxG	1-294/294	3/7	97.96	96.27	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_152574	136	424	-	esxG	3-294/294	3/5	97.96	95.22	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_79966	936	1227	+	esxG	1-294/294	7/10	97.96	93.96	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_63188	195	479	-	esxH	1-289/291	3/4	97.94	96.89	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_223794	165	451	-	esxH	1-289/291	5/6	97.94	95.19	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_35503	456	743	-	esxH	1-291/291	5/9	97.94	80.27	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_26748	2122	2404	+	esxA	1-288/288	5/7	97.92	94.81	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_90337	13	299	-	esxA	1-288/288	6/11	97.92	92.15	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_28535	282	562	+	esxN	2-285/285	4/7	97.89	94.41	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_43593	203	484	+	esxN	1-279/285	1/3	97.89	92.55	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_175226	22	300	+	esxN	1-283/285	4/4	97.89	92.23	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_209953	280	560	+	esxN	1-285/285	6/8	97.89	85.02	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_54160	180	465	-	esxN	1-283/285	8/11	97.89	84.48	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_220974	30	1176	+	mbtN	1-1161/1161	21/36	97.85	94.2	NP_215862	(mbtN) Acyl-CoA dehydrogenase MbtN [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_155601	563	1387	+	lipF	1-833/834	14/26	97.84	94.89	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_142746	672	989	+	mbtL	1-318/321	5/8	97.82	90.37	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_88085	360	1203	+	erp	1-855/855	17/27	97.78	94.44	NP_218327	(erp) exported repetitive protein precursor Erp [Erp (VF0312)] [Mycobacterium tuberculosis H37Rv]	VF0312
k141_137391	803	1392	-	PPE41	1-585/585	20/31	97.78	90.55	YP_177881	(PPE41) PPE family protein PPE41 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_232498	216	512	+	esxB	7-303/303	2/2	97.69	98.99	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_211546	271	567	+	esxB	1-303/303	6/8	97.69	96.71	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_116899	175	390	+	mbtH	1-215/216	5/9	97.69	95.0	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_176012	91	381	+	esxM	1-294/297	4/5	97.64	93.56	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_108091	210	500	-	esxM	1-296/297	5/7	97.64	91.25	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_27522	271	562	-	esxM	1-296/297	3/8	97.64	90.94	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_26204	867	2235	+	eccE1	1-1389/1389	28/46	97.62	94.22	NP_218399	(eccE1) ESX-1 type VII secretion system EccE1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_175588	862	1150	+	esxG	1-294/294	8/9	97.62	93.24	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_55742	939	2296	+	espB	1-1380/1383	26/38	97.61	93.37	NP_218398	(espB) ESX-1 type VII secretion system EspB [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_155334	171	455	-	esxH	1-291/291	5/8	97.59	95.89	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_141881	1408	2383	+	eccE3	1-996/996	19/28	97.59	94.2	NP_214806	(eccE3) Type VII secretion system transmembrane protein EccE3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_49981	224	507	+	esxH	1-288/291	2/4	97.59	94.1	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_27781	177	456	+	esxN	1-283/285	4/7	97.54	93.33	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_73408	129	407	-	esxN	1-279/285	2/2	97.54	91.43	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_117885	189	468	+	esxN	1-283/285	4/7	97.54	89.47	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_65731	634	911	-	esxN	1-285/285	3/7	97.54	89.12	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_225803	203	484	+	esxN	2-285/285	8/10	97.54	88.54	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_76055	239	518	+	esxN	1-279/285	3/3	97.54	81.5	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_123052	62	377	-	mbtL	6-321/321	4/6	97.51	96.55	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_239438	39	352	+	mbtL	1-321/321	7/9	97.51	93.79	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_27701	131	1550	+	eccB1	3-1443/1443	29/47	97.51	91.88	NP_218386	(eccB1) ESX-1 type VII secretion system protein EccB1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_81322	581	1470	-	Rv1794	1-903/903	18/33	97.45	91.35	NP_216310	(Rv1794) ESX-5 locus protein [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_130349	192	497	-	espC	1-307/312	4/5	97.44	94.17	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_211073	243	543	+	PE5	1-308/309	5/7	97.41	97.08	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_143743	1	292	-	PE35	1-293/300	1/1	97.33	99.32	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_199978	33	322	+	esxM	1-290/297	2/2	97.31	97.25	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_143806	4921	5659	+	phoP	1-744/744	26/35	97.31	92.36	NP_215271	(phoP) Possible two component system response transcriptional positive regulator PhoP [PhoP (VF0286)] [Mycobacterium tuberculosis H37Rv]	VF0286
k141_214375	329	622	-	esxM	1-295/297	6/11	97.31	91.67	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_208759	408	1571	-	espA	1-1179/1179	31/49	97.29	92.98	NP_218133	(espA) ESX-1 type VII secretion system EspA [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_1116	802	1088	+	esxG	1-290/294	4/5	97.28	97.25	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_243736	656	1659	+	fbpC	1-1022/1023	18/36	97.26	93.89	YP_177694	(fbpC) Secreted antigen 85-C FbpC (85C) (antigen 85 complex C) (AG58C) (mycolyl transferase 85C) (fibronectin-binding protein C) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_154386	1459	1742	-	esxH	1-288/291	4/6	97.25	96.19	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_91282	508	939	-	hspX	1-435/435	8/21	97.24	89.41	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301
k141_246767	1786	2063	-	esxN	1-283/285	4/7	97.19	89.79	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_97840	65	682	+	mbtK	1-632/633	9/20	97.16	95.59	NP_215863	(mbtK) Lysine N-acetyltransferase MbtK [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_149381	173	790	-	mbtK	1-633/633	9/21	97.16	94.97	NP_215863	(mbtK) Lysine N-acetyltransferase MbtK [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_155334	823	2336	-	PPE4	1-1542/1542	33/60	97.15	94.22	YP_177711	(PPE4) PPE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_169710	66	1267	-	eccE5	7-1221/1221	18/45	97.13	93.75	NP_216313	(eccE5) ESX-5 type VII secretion system EccE5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_182225	269	812	-	espD	9-554/555	7/12	97.12	96.37	NP_218131	(espD) ESX-1 type VII secretion system EspD [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_14339	654	2383	-	mycP5	1-1758/1758	38/74	97.1	91.58	NP_216312	(mycP5) ESX-5 type VII secretion system subtilisin-like serine protease MycP5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_9822	51	350	-	PE5	1-309/309	3/9	97.09	96.76	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_81004	414	1255	+	erp	1-855/855	16/37	97.08	93.54	NP_218327	(erp) exported repetitive protein precursor Erp [Erp (VF0312)] [Mycobacterium tuberculosis H37Rv]	VF0312
k141_259918	797	2285	+	eccB5	1-1521/1521	39/58	97.04	92.11	NP_216298	(eccB5) ESX-5 type VII secretion system protein EccB5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_194689	269	562	-	esxB	1-303/303	4/9	97.03	97.03	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_198122	668	962	+	esxB	1-300/303	4/7	97.03	97.01	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_161117	1	298	+	esxB	7-303/303	4/7	97.03	95.35	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_165429	541	1226	+	mgtC	1-705/705	11/23	97.02	94.2	NP_216327	(mgtC) Possible Mg2+ transport P-type ATPase C MgtC [MgtC (VF0289)] [Mycobacterium tuberculosis H37Rv]	VF0289
k141_103154	1805	2386	-	hbhA	1-600/600	1/18	97.0	97.0	NP_214989	(hbhA) iron-regulated heparin binding hemagglutinin hbhA (adhesin) [HbhA (VF0313)] [Mycobacterium tuberculosis H37Rv]	VF0313
k141_31177	588	1865	+	mbtG	1-1296/1296	33/60	96.99	91.65	NP_216894	(mbtG) Lysine-N-oxygenase MbtG (L-lysine 6-monooxygenase) (lysine N6-hydroxylase) [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_75844	112	1107	+	fbpC	1-994/1023	5/6	96.97	98.1	YP_177694	(fbpC) Secreted antigen 85-C FbpC (85C) (antigen 85 complex C) (AG58C) (mycolyl transferase 85C) (fibronectin-binding protein C) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_65474	289	577	-	esxM	1-295/297	4/8	96.97	93.58	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_90741	865	2063	+	eccE5	1-1221/1221	31/52	96.97	90.13	NP_216313	(eccE5) ESX-5 type VII secretion system EccE5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_87040	214	1048	-	erp	1-855/855	18/32	96.96	92.45	NP_218327	(erp) exported repetitive protein precursor Erp [Erp (VF0312)] [Mycobacterium tuberculosis H37Rv]	VF0312
k141_166693	636	2379	+	eccCb1	1-1776/1776	50/76	96.96	91.66	NP_218388	(eccCb1) ESX-1 type VII secretion system FtsK/SpoIIIE family protein EccCb1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_51341	103	390	-	esxG	1-290/294	5/8	96.94	83.62	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_29433	1127	1411	+	esxH	1-291/291	7/12	96.91	92.52	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_186934	476	1842	-	mycP3	1-1386/1386	41/67	96.9	89.57	NP_214805	(mycP3) Type VII secretion system membrane-anchored mycosin MycP3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_176020	613	923	-	mbtL	1-321/321	3/10	96.88	94.08	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_168331	42	325	+	esxA	1-288/288	6/14	96.88	91.81	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_161970	267	542	-	esxN	1-283/285	3/7	96.84	93.64	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_140247	114	390	+	esxN	1-279/285	3/4	96.84	93.21	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_110884	521	796	+	esxN	4-285/285	2/6	96.84	89.36	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_207011	647	2041	-	eccD3	1-1419/1419	40/68	96.76	90.98	NP_214804	(eccD3) Type IIV secretion system transmembrane protein EccD3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_35290	589	890	+	esxB	1-299/303	10/15	96.7	90.58	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_107971	434	1825	+	eccD3	1-1419/1419	36/67	96.69	91.17	NP_214804	(eccD3) Type IIV secretion system transmembrane protein EccD3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_129366	439	1971	-	mbtM	7-1563/1566	36/62	96.68	92.64	NP_215861	(mbtM) Probable fatty acyl-AMP ligase MbtM [FadD33 (VF0308)] [Mycobacterium tuberculosis H37Rv]	VF0308
k141_120868	237	533	-	PE35	6-300/300	6/12	96.67	93.38	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_63779	96	683	+	hbhA	1-600/600	14/28	96.67	91.78	NP_214989	(hbhA) iron-regulated heparin binding hemagglutinin hbhA (adhesin) [HbhA (VF0313)] [Mycobacterium tuberculosis H37Rv]	VF0313
k141_138602	643	940	+	PE35	1-300/300	13/18	96.67	85.06	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_113421	708	1524	+	lipF	1-834/834	22/39	96.64	90.3	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307
k141_110884	203	494	+	esxM	1-295/297	6/13	96.63	89.67	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_24991	166	449	-	esxG	5-294/294	3/6	96.6	96.9	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_52625	315	605	+	esxG	1-291/294	6/14	96.6	93.29	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_258618	1169	1454	+	esxG	3-294/294	5/10	96.6	92.86	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_95602	267	556	+	esxG	1-294/294	9/16	96.6	86.33	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_170063	987	1277	+	esxG	1-290/294	3/13	96.6	80.81	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_9857	61	375	+	mbtL	1-321/321	10/16	96.57	90.8	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_243184	1	281	+	esxH	11-291/291	0/0	96.56	99.29	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_39883	3047	3329	+	esxH	1-287/291	5/8	96.56	80.28	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_83248	453	728	-	esxN	1-278/285	3/4	96.49	96.42	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_27811	173	447	+	esxN	1-279/285	3/4	96.49	96.06	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_196439	391	670	-	esxN	1-285/285	9/15	96.49	88.97	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_142712	22	322	-	espC	9-312/312	3/3	96.47	99.01	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_190178	50	1200	+	espA	1-1175/1179	31/52	96.44	91.42	NP_218133	(espA) ESX-1 type VII secretion system EspA [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_105241	125	1969	-	eccA3	1-1896/1896	45/85	96.41	91.79	NP_214796	(eccA3) Type VII secretion system protein EccA3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_154386	582	1443	-	espG3	1-888/888	20/38	96.4	92.51	NP_214803	(espG3) Type VII secretion system protein EspG3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_165230	435	734	-	esxB	1-303/303	9/19	96.37	91.64	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_172109	174	463	+	PE35	1-291/300	3/3	96.33	97.94	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_22605	1345	1930	-	hbhA	1-600/600	17/30	96.33	91.12	NP_214989	(hbhA) iron-regulated heparin binding hemagglutinin hbhA (adhesin) [HbhA (VF0313)] [Mycobacterium tuberculosis H37Rv]	VF0313
k141_26748	276	565	+	PE35	1-300/300	5/12	96.33	89.04	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_159130	686	1116	+	hspX	9-434/435	10/19	96.32	91.78	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_256103	281	489	-	mbtH	1-210/216	3/3	96.3	95.26	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_159550	1266	1576	-	mbtL	12-320/321	1/2	96.26	94.21	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_78828	370	1038	+	ideR	17-693/693	8/12	96.25	96.61	NP_217227	(ideR) Iron-dependent repressor and activator IdeR [IdeR (VF0300)] [Mycobacterium tuberculosis H37Rv]	VF0300
k141_26291	33	1527	+	eccB5	1-1521/1521	48/88	96.25	88.79	NP_216298	(eccB5) ESX-5 type VII secretion system protein EccB5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_199170	660	1227	-	PPE41	1-585/585	17/27	96.24	92.03	YP_177881	(PPE41) PPE family protein PPE41 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_73230	307	1962	-	mbtA	5-1698/1698	41/82	96.23	91.14	NP_216900	(mbtA) Bifunctional enzyme MbtA: salicyl-AMP ligase (SAL-AMP ligase) + salicyl-S-ArCP synthetase [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_27037	2868	3409	+	espD	4-555/555	16/26	96.22	91.61	NP_218131	(espD) ESX-1 type VII secretion system EspD [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_95129	690	1271	+	hbhA	10-598/600	9/17	96.17	95.12	NP_214989	(hbhA) iron-regulated heparin binding hemagglutinin hbhA (adhesin) [HbhA (VF0313)] [Mycobacterium tuberculosis H37Rv]	VF0313
k141_208787	37	343	-	espC	3-312/312	11/17	96.15	88.33	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_122578	321	594	-	esxN	1-274/285	0/0	96.14	98.54	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_128109	206	480	-	esxN	1-281/285	5/8	96.14	91.49	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_174795	6747	7024	+	esxN	1-283/285	8/13	96.14	85.37	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_142155	113	659	+	espD	4-552/555	20/30	96.04	88.1	NP_218131	(espD) ESX-1 type VII secretion system EspD [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_198802	37	847	+	lipF	1-834/834	22/43	96.04	86.49	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307
k141_129786	1	295	+	PE35	13-300/300	5/7	96.0	94.58	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_211262	457	1038	+	hbhA	1-600/600	15/30	96.0	92.41	NP_214989	(hbhA) iron-regulated heparin binding hemagglutinin hbhA (adhesin) [HbhA (VF0313)] [Mycobacterium tuberculosis H37Rv]	VF0313
k141_166693	2519	2817	+	PE35	1-300/300	11/23	96.0	86.5	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_222472	2489	3484	+	fbpA	1-1017/1017	33/61	95.97	89.01	NP_218321	(fbpA) Secreted antigen 85-a FbpA (mycolyl transferase 85A) (fibronectin-binding protein A) (antigen 85 complex A) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_13920	1	285	+	esxM	12-297/297	1/1	95.96	98.25	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_114999	1	285	-	esxM	1-285/297	0/0	95.96	97.54	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_183332	1	285	+	esxM	11-296/297	1/1	95.96	96.5	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_163056	748	1041	-	esxM	1-297/297	13/21	95.96	85.95	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_75791	578	887	+	mbtL	1-317/321	5/11	95.95	93.73	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_251074	319	1136	-	lipF	3-834/834	23/50	95.92	89.18	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_32620	568	2044	+	eccB5	1-1521/1521	39/82	95.86	90.71	NP_216298	(eccB5) ESX-5 type VII secretion system protein EccB5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_137286	713	2422	-	mycP5	8-1758/1758	49/91	95.85	90.03	NP_216312	(mycP5) ESX-5 type VII secretion system subtilisin-like serine protease MycP5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_31177	1843	2053	+	mbtH	1-208/216	4/5	95.83	95.75	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_22620	287	589	+	espC	1-312/312	9/17	95.83	91.77	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_243358	1	297	-	PE5	1-297/309	2/2	95.79	98.99	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_165956	1484	1762	+	esxN	10-283/285	5/7	95.79	91.07	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_187270	2132	3023	-	Rv1794	1-903/903	39/65	95.79	88.39	NP_216310	(Rv1794) ESX-5 locus protein [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_116509	523	1414	-	mbtJ	3-921/921	27/47	95.77	91.28	YP_177876	(mbtJ) Putative acetyl hydrolase MbtJ [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_27878	1265	1947	+	mgtC	1-705/705	20/38	95.74	87.66	NP_216327	(mgtC) Possible Mg2+ transport P-type ATPase C MgtC [MgtC (VF0289)] [Mycobacterium tuberculosis H37Rv]	VF0289
k141_214492	842	2155	+	mbtI	1-1353/1353	38/79	95.64	89.51	YP_177877	(mbtI) Isochorismate synthase MbtI [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_13935	55	340	-	esxM	12-295/297	1/2	95.62	96.85	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_53901	419	703	+	esxM	1-295/297	5/12	95.62	92.23	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_145713	42	716	+	mgtC	1-701/705	13/30	95.46	91.75	NP_216327	(mgtC) Possible Mg2+ transport P-type ATPase C MgtC [MgtC (VF0289)] [Mycobacterium tuberculosis H37Rv]	VF0289
k141_186602	700	974	-	esxN	1-276/285	4/7	95.44	92.11	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_45034	173	444	-	esxN	10-284/285	3/3	95.44	82.55	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_37665	934	1557	+	mbtK	8-633/633	22/42	95.42	87.62	NP_215863	(mbtK) Lysine N-acetyltransferase MbtK [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_61677	482	765	-	esxM	1-295/297	7/13	95.29	90.88	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_156637	19	313	-	esxM	1-295/297	9/24	95.29	86.32	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_226610	472	2241	-	eccA5	1-1833/1833	49/111	95.25	91.17	NP_216314	(eccA5) ESX-5 type VII secretion system AAA+ ATPase EccA5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_188258	242	1578	-	espB	1-1383/1383	43/86	95.23	88.17	NP_218398	(espB) ESX-1 type VII secretion system EspB [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_245078	213	1021	-	lipF	1-834/834	26/55	95.2	90.46	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307
k141_214375	1	276	-	esxN	1-272/285	2/6	95.09	94.95	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_51267	320	592	+	esxN	1-279/285	6/10	95.09	91.81	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_60803	134	989	+	espG3	1-882/888	25/50	95.05	89.93	NP_214803	(espG3) Type VII secretion system protein EspG3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_31772	218	525	+	mbtL	1-317/321	8/15	95.02	92.5	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_86112	473	1051	+	hbhA	7-600/600	24/33	95.0	89.22	NP_214989	(hbhA) iron-regulated heparin binding hemagglutinin hbhA (adhesin) [HbhA (VF0313)] [Mycobacterium tuberculosis H37Rv]	VF0313
k141_131221	1688	1976	+	esxM	1-297/297	12/22	94.95	84.21	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_173006	422	1711	-	mycP1	36-1341/1341	29/50	94.93	91.69	NP_218400	(mycP1) ESX-1 type VII secretion system subtilisin-like serine protease MycP1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_161484	1220	1430	-	mbtH	1-205/216	4/6	94.91	93.84	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_44297	740	2219	-	eccD1	2-1531/1536	49/96	94.86	86.99	NP_218394	(eccD1) ESX-1 type VII secretion system EccD1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_122097	94	389	+	PE5	1-304/309	7/14	94.82	93.81	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_156166	56	1315	-	eccCa5	1-1306/1308	46/86	94.8	89.22	NP_216299	(eccCa5) ESX-5 type VII secretion system FtsK/SpoIIIE family protein EccCa5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_41070	1	1352	-	eccD3	26-1407/1419	21/44	94.79	94.31	NP_214804	(eccD3) Type IIV secretion system transmembrane protein EccD3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_240868	24	1319	+	mbtC	1-1332/1335	57/98	94.76	86.28	NP_216898	(mbtC) Polyketide synthetase MbtC (polyketide synthase) [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_154544	150	423	-	esxN	12-285/285	6/8	94.74	95.32	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_82564	2	271	+	esxN	13-282/285	0/0	94.74	92.96	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_124366	1	270	+	esxN	13-285/285	1/3	94.74	92.67	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_25652	3300	3574	+	esxN	1-285/285	9/20	94.74	88.28	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_230300	1	287	+	esxB	13-303/303	2/4	94.72	97.59	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_40127	260	1063	-	lipF	1-832/834	21/56	94.72	89.01	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307
k141_104306	1477	1763	+	esxM	1-295/297	11/20	94.61	87.71	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_231753	3104	3388	-	esxM	1-295/297	15/18	94.61	85.28	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_95950	1399	1680	-	esxM	9-295/297	5/7	94.61	81.25	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_27251	157	1287	+	espA	1-1179/1179	50/80	94.57	84.6	NP_218133	(espA) ESX-1 type VII secretion system EspA [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_223794	481	761	-	esxG	14-294/294	4/6	94.56	96.83	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_236266	202	1657	+	eccB5	8-1500/1521	41/73	94.54	92.79	NP_216298	(eccB5) ESX-5 type VII secretion system protein EccB5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_243510	1849	2140	+	PE5	1-298/309	3/6	94.5	97.31	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_34920	64	360	+	PE5	1-308/309	10/21	94.5	90.73	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_229096	1044	1715	-	mgtC	14-696/705	11/23	94.47	94.34	NP_216327	(mgtC) Possible Mg2+ transport P-type ATPase C MgtC [MgtC (VF0289)] [Mycobacterium tuberculosis H37Rv]	VF0289
k141_139013	1190	3350	-	espK	9-2185/2190	77/202	94.43	86.87	NP_218396	(espK) ESX-1 type VII secretion system EspK [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_113	184	1864	+	irtB	1-1740/1740	78/135	94.43	83.63	NP_215865	(irtB) Iron-regulated transporter IrtB [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_168555	290	1145	-	Rv1794	22-903/903	21/34	94.35	91.99	NP_216310	(Rv1794) ESX-5 locus protein [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_77004	123	404	+	esxM	1-287/297	5/9	94.28	92.04	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_187270	3422	3702	-	esxM	1-297/297	11/18	94.28	88.59	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_177029	183	453	+	esxN	7-274/285	1/3	94.04	94.83	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_11330	357	628	-	esxN	1-268/285	1/4	94.04	93.38	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_226654	580	862	-	PE35	6-293/300	4/7	94.0	97.23	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_25652	2963	3247	+	esxM	6-288/297	5/10	93.94	93.08	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_82488	51	326	-	esxG	13-290/294	1/2	93.88	84.89	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_209202	447	1012	+	hbhA	1-573/600	10/15	93.67	90.64	NP_214989	(hbhA) iron-regulated heparin binding hemagglutinin hbhA (adhesin) [HbhA (VF0313)] [Mycobacterium tuberculosis H37Rv]	VF0313

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_73408	459	736	-	esxM	2-296/297	7/17	93.6	89.49	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_27037	2444	2742	+	espC	1-306/312	12/21	93.59	84.34	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_221532	1	407	+	hspX	16-435/435	4/13	93.56	95.95	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301
k141_50930	482	784	-	mbtL	5-318/321	9/17	93.46	90.85	NP_215860	(mbtL) Acyl carrier protein (ACP) MbtL [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_181899	1	284	-	esxB	1-283/303	1/1	93.4	98.59	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_42557	1	291	-	esxB	1-292/303	7/17	93.4	90.33	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_146682	3346	4266	-	fbpB	1-958/978	20/53	93.35	92.86	NP_216402	(fbpB) Secreted antigen 85-B FbpB (85B) (antigen 85 complex B) (mycolyl transferase 85B) (fibronectin-binding protein B) (extracellular alpha-antigen) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_51147	1	266	+	esxN	13-283/285	3/5	93.33	97.05	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_87293	2417	2830	+	hspX	16-435/435	14/22	93.33	90.19	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301
k141_115654	15	293	+	esxM	1-277/297	1/2	93.27	96.06	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_208759	37	331	-	espC	12-311/312	8/13	93.27	91.45	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_196439	722	1000	-	esxM	3-295/297	10/18	93.27	89.83	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_13126	429	1870	+	eccD1	1-1534/1536	44/114	93.16	88.03	NP_218394	(eccD1) ESX-1 type VII secretion system EccD1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_96128	2291	3389	-	espA	80-1179/1179	3/3	93.13	99.64	NP_218133	(espA) ESX-1 type VII secretion system EspA [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_20443	1	1505	-	eccB3	3-1514/1617	3/7	93.07	98.61	NP_214797	(eccB3) Type VII secretion system protein EccB3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_214492	2167	3045	-	mbtJ	1-909/921	35/74	93.05	84.86	YP_177876	(mbtJ) Putative acetyl hydrolase MbtJ [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_72720	3	268	+	esxN	16-281/285	2/2	92.98	82.4	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_75070	412	1942	+	eccB3	1-1616/1617	69/141	92.95	85.4	NP_214797	(eccB3) Type VII secretion system protein EccB3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_182229	495	770	-	esxM	21-296/297	0/0	92.93	97.46	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_11330	682	960	-	esxM	19-294/297	2/3	92.93	94.98	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_237911	766	2086	+	eccD3	1-1333/1419	12/18	92.88	96.78	NP_214804	(eccD3) Type IIV secretion system transmembrane protein EccD3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_202562	162	1118	+	fbpC	1-987/1023	21/44	92.86	91.15	YP_177694	(fbpC) Secreted antigen 85-C FbpC (85C) (antigen 85 complex C) (AG58C) (mycolyl transferase 85C) (fibronectin-binding protein C) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_6609	866	1650	+	lipF	5-831/834	28/64	92.81	87.11	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_198927	1003	1283	+	esxB	7-303/303	6/16	92.74	93.27	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_255290	1	272	-	esxA	1-272/288	8/10	92.71	90.61	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_143921	71	339	+	esxN	1-273/285	5/14	92.63	87.77	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_110652	1	275	-	esxM	1-277/297	2/2	92.59	96.03	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_174795	1935	2210	-	esxM	5-295/297	9/17	92.59	89.38	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_99094	1502	1781	+	esxM	1-295/297	17/25	92.59	83.33	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_239870	1355	2781	-	eccB5	1-1521/1521	67/132	92.57	81.36	NP_216298	(eccB5) ESX-5 type VII secretion system protein EccB5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_80979	1	925	-	eccE3	3-930/996	10/11	92.47	97.21	NP_214806	(eccE3) Type VII secretion system transmembrane protein EccE3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_168704	1	283	+	esxB	18-303/303	7/9	92.41	94.12	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_76939	2756	4464	-	eccA5	89-1833/1833	33/68	92.36	93.81	NP_216314	(eccA5) ESX-5 type VII secretion system AAA+ ATPase EccA5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_112556	608	874	-	esxN	1-266/285	6/7	92.28	80.74	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_209843	1	1401	+	eccD5	89-1512/1512	18/35	92.26	95.94	NP_216311	(eccD5) ESX-5 type VII secretion system EccD5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_30070	1	274	-	esxM	10-284/297	1/1	92.26	95.27	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_192828	340	1243	-	fbpB	67-978/978	8/16	92.02	97.27	NP_216402	(fbpB) Secreted antigen 85-B FbpB (85B) (antigen 85 complex B) (mycolyl transferase 85B) (fibronectin-binding protein B) (extracellular alpha-antigen) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_226125	2415	2676	-	esxN	1-265/285	2/3	91.93	92.08	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_65421	1	263	-	esxN	1-263/285	2/2	91.93	91.29	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_174745	9	273	-	esxN	20-285/285	6/7	91.93	89.96	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_143914	1	262	-	esxN	1-273/285	2/11	91.93	88.28	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_88544	663	925	-	esxN	10-285/285	9/15	91.93	82.31	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_257139	453	725	-	esxM	19-296/297	3/5	91.92	95.32	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_81124	82	355	-	esxM	17-296/297	4/8	91.92	93.59	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_190907	2487	2765	-	esxM	1-286/297	11/19	91.92	89.73	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_65488	1	261	-	esxN	1-263/285	1/2	91.58	95.44	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_89031	1	263	+	esxN	13-273/285	1/2	91.58	92.02	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_196356	1	769	-	lipF	1-773/834	9/16	91.49	96.41	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307
k141_143806	750	1446	-	phoP	1-744/744	29/81	91.4	84.62	NP_215271	(phoP) Possible two component system response transcriptional positive regulator PhoP [PhoP (VF0286)] [Mycobacterium tuberculosis H37Rv]	VF0286
k141_96772	1	840	+	Rv1794	70-903/903	13/24	91.36	93.17	NP_216310	(Rv1794) ESX-5 locus protein [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_241674	1	1304	+	eccD3	22-1333/1419	17/26	91.26	95.69	NP_214804	(eccD3) Type IIV secretion system transmembrane protein EccD3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_247196	3	274	-	esxM	13-284/297	2/2	91.25	83.15	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_234029	98	357	+	esxN	1-260/285	0/0	91.23	93.08	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_39471	337	904	-	hbhA	24-578/600	20/29	91.17	89.24	NP_214989	(hbhA) iron-regulated heparin binding hemagglutinin hbhA (adhesin) [HbhA (VF0313)] [Mycobacterium tuberculosis H37Rv]	VF0313
k141_39883	2729	3003	+	esxG	1-275/294	11/14	91.16	80.5	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_172176	1	397	-	hspX	1-410/435	9/17	90.8	92.96	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301
k141_213218	1	813	+	espG3	17-834/888	11/19	90.77	96.36	NP_214803	(espG3) Type VII secretion system protein EspG3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_174783	850	1045	+	mbtH	1-199/216	1/3	90.74	97.49	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_86308	305	577	-	PE35	26-300/300	3/4	90.67	97.46	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_16651	1	778	-	erp	1-787/855	11/15	90.64	96.46	NP_218327	(erp) exported repetitive protein precursor Erp [Erp (VF0312)] [Mycobacterium tuberculosis H37Rv]	VF0312
k141_51267	1	273	+	esxM	22-297/297	4/11	90.57	92.14	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_174795	6448	6718	+	esxM	1-295/297	10/28	90.57	85.52	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_216100	558	823	+	esxG	1-267/294	1/1	90.48	98.88	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_253735	1027	1292	-	esxG	16-290/294	3/9	90.48	83.27	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_187270	479	1876	-	eccD5	3-1512/1512	64/172	90.48	81.69	NP_216311	(eccD5) ESX-5 type VII secretion system EccD5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_250495	1	1342	+	phoR	71-1448/1458	48/82	90.47	87.51	NP_215272	(phoR) Possible two component system response sensor kinase membrane associated PhoR [PhoP/R (CVF331)] [Mycobacterium tuberculosis H37Rv]	NA
k141_92283	177	374	+	mbtH	9-203/216	1/3	90.28	98.48	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_88544	980	1252	-	esxM	1-276/297	8/13	90.24	89.32	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_28547	1	257	-	esxN	1-263/285	3/6	90.18	95.06	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_72654	104	360	+	esxN	1-259/285	1/2	90.18	91.51	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_180218	121	379	+	esxN	1-261/285	5/6	90.18	88.21	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_93859	1049	1308	-	esxN	1-266/285	8/12	90.18	85.87	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_138384	92	352	-	esxN	22-285/285	6/11	90.18	85.82	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_101230	784	1967	-	eccCa5	79-1295/1308	23/43	90.14	92.72	NP_216299	(eccCa5) ESX-5 type VII secretion system FtsK/SpoIIIE family protein EccCa5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_61465	1	782	+	erp	71-855/855	18/29	89.94	91.48	NP_218327	(erp) exported repetitive protein precursor Erp [Erp (VF0312)] [Mycobacterium tuberculosis H37Rv]	VF0312
k141_51276	1	268	+	esxM	22-295/297	3/8	89.9	93.82	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_234504	224	479	+	esxN	24-285/285	3/6	89.82	90.08	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_153821	128	321	-	mbtH	18-216/216	3/5	89.81	88.44	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_107878	1	269	+	PE35	29-298/300	1/1	89.67	98.15	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_234771	959	1224	+	esxM	21-296/297	7/10	89.56	87.32	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_202309	1	266	-	esxM	13-279/297	1/1	89.56	81.27	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_234274	6	266	-	esxH	1-260/291	1/1	89.35	83.53	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_16844	1	764	+	lipF	69-834/834	20/40	89.33	91.59	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307
k141_143249	111	381	+	esxB	1-272/303	3/3	89.11	92.67	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_97463	1	264	-	esxM	1-265/297	1/1	88.89	96.23	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_47993	139	403	-	esxM	31-295/297	2/2	88.89	95.49	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_36707	165	430	+	esxM	1-277/297	2/15	88.89	91.04	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_52235	1	253	-	esxN	1-257/285	2/4	88.77	93.0	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_120404	1	254	-	esxN	1-257/285	4/5	88.77	80.23	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_51147	354	1161	+	Rv1794	1-894/903	27/100	88.7	85.91	NP_216310	(Rv1794) ESX-5 locus protein [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_29080	1	263	-	esxM	1-265/297	2/2	88.55	96.23	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_54629	1	263	-	esxM	1-264/297	1/1	88.55	95.45	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_142007	1	764	-	erp	82-843/855	9/12	88.54	96.88	NP_218327	(erp) exported repetitive protein precursor Erp [Erp (VF0312)] [Mycobacterium tuberculosis H37Rv]	VF0312
k141_108135	1	264	-	esxN	1-253/285	2/13	88.42	88.68	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_190907	2148	2408	-	esxN	21-283/285	12/20	88.42	82.35	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_93821	1258	1883	+	mgtC	1-635/705	9/15	88.37	95.45	NP_216327	(mgtC) Possible Mg2+ transport P-type ATPase C MgtC [MgtC (VF0289)] [Mycobacterium tuberculosis H37Rv]	VF0289
k141_25747	1	277	-	PE5	1-274/309	3/5	88.35	96.04	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_80890	249	517	-	PE35	34-300/300	4/6	88.33	95.94	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_210662	149	418	+	PE35	1-269/300	4/9	88.33	95.62	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_156152	3	1446	+	eccB3	1-1469/1617	29/57	88.31	92.32	NP_214797	(eccB3) Type VII secretion system protein EccB3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_21573	1	802	-	espG3	1-809/888	25/43	88.29	91.17	NP_214803	(espG3) Type VII secretion system protein EspG3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_241071	3268	3530	-	esxG	21-281/294	4/6	88.1	83.77	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_174795	1628	1880	-	esxN	23-285/285	9/14	88.07	85.66	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_185318	730	980	-	esxN	1-257/285	3/6	88.07	80.93	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_135307	58	313	-	esxH	29-291/291	2/7	87.97	82.13	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_98821	1	261	-	esxM	2-265/297	1/3	87.88	81.82	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_182190	1	1181	-	mycP1	1-1210/1341	20/35	87.84	94.72	NP_218400	(mycP1) ESX-1 type VII secretion system subtilisin-like serine protease MycP1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_65474	1	250	-	esxN	10-263/285	3/4	87.72	91.34	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_187270	3124	3373	-	esxN	1-273/285	9/23	87.72	86.45	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_104306	1815	2069	+	esxN	24-282/285	8/14	87.72	81.82	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_90610	1	277	-	espC	1-274/312	4/5	87.5	97.48	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_99094	1832	2085	+	esxN	27-280/285	5/10	87.37	83.78	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_149962	1	813	-	mbtJ	1-816/921	12/21	87.3	94.79	YP_177876	(mbtJ) Putative acetyl hydrolase MbtJ [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_95602	588	845	+	esxH	1-255/291	4/5	87.29	96.14	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_69400	324	580	-	esxH	16-273/291	6/7	87.29	92.72	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_3987	769	956	-	mbtH	24-216/216	3/5	87.04	96.37	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_145129	325	1069	-	erp	103-850/855	4/5	87.02	97.86	NP_218327	(erp) exported repetitive protein precursor Erp [Erp (VF0312)] [Mycobacterium tuberculosis H37Rv]	VF0312
k141_52768	193	440	-	esxN	1-256/285	2/8	87.02	80.86	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_210718	409	794	-	hspX	4-386/435	6/13	86.9	94.63	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301
k141_89566	252	1513	-	eccB1	169-1440/1443	20/28	86.83	95.39	NP_218386	(eccB1) ESX-1 type VII secretion system protein EccB1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_211102	166	428	+	esxB	1-272/303	4/9	86.8	93.02	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_136498	1	749	+	erp	103-855/855	14/18	86.78	95.92	NP_218327	(erp) exported repetitive protein precursor Erp [Erp (VF0312)] [Mycobacterium tuberculosis H37Rv]	VF0312
k141_220961	31	1449	+	eccB3	1-1437/1617	31/50	86.77	92.36	NP_214797	(eccB3) Type VII secretion system protein EccB3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_10985	297	565	+	PE5	1-269/309	2/2	86.73	98.89	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_192616	620	896	-	PE5	33-309/309	9/18	86.73	92.31	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_18757	1	257	+	esxG	29-287/294	5/6	86.73	84.67	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_132472	570	1780	-	mycP3	163-1386/1386	18/31	86.72	95.86	NP_214805	(mycP3) Type VII secretion system membrane-anchored mycosin MycP3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_7089	1	260	+	PE35	39-300/300	1/2	86.67	98.09	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_199354	1	257	-	esxM	1-257/297	0/0	86.53	97.67	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_66107	291	547	+	esxG	1-258/294	4/7	86.39	95.02	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_36339	130	383	-	esxG	10-263/294	0/0	86.39	85.43	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_38013	707	1879	-	mbtI	155-1353/1353	21/38	86.25	94.36	YP_177877	(mbtI) Isochorismate synthase MbtI [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_226730	505	1396	+	fbpC	1-886/1023	12/18	86.02	96.21	YP_177694	(fbpC) Secreted antigen 85-C FbpC (85C) (antigen 85 complex C) (AG58C) (mycolyl transferase 85C) (fibronectin-binding protein C) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_56533	1	261	+	PE35	39-300/300	5/7	86.0	95.85	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_2215	104	359	+	esxM	1-257/297	2/3	85.86	94.96	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_197686	1	257	+	PE35	32-293/300	3/5	85.67	98.09	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_107689	1	185	-	mbtH	1-185/216	0/0	85.65	99.46	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_41215	125	371	-	esxN	38-283/285	2/5	85.61	91.57	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_174872	1190	3131	+	eccCa1	1-1960/2244	47/68	85.43	93.75	NP_218387	(eccCa1) ESX-1 type VII secretion system FtsK/SpoIIIE family protein EccCa1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_212790	1	257	-	esxG	1-251/294	4/6	85.37	96.11	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_5617	1	255	-	esxG	1-254/294	5/7	85.37	95.74	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_87194	1	371	-	hspX	1-379/435	4/8	85.29	96.57	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_162342	1	243	+	esxN	36-281/285	1/3	85.26	91.87	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_162362	398	642	-	esxN	37-281/285	3/4	85.26	91.09	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_191217	437	685	-	esxH	44-291/291	1/1	85.22	83.53	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_151568	1	248	-	esxH	16-269/291	3/6	85.22	80.71	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_93859	1364	1616	-	esxM	22-294/297	9/20	85.19	85.35	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_110398	1	258	+	esxB	39-303/303	4/7	85.15	97.36	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_78482	1	541	-	mbtK	1-549/633	7/12	85.15	96.01	NP_215863	(mbtK) Lysine N-acetyltransferase MbtK [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_51383	1	1182	-	eccE1	15-1214/1389	9/18	85.1	96.83	NP_218399	(eccE1) ESX-1 type VII secretion system EccE1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_22525	1	246	-	esxA	1-249/288	3/5	85.07	97.2	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_100850	1	256	-	PE35	1-256/300	2/2	85.0	97.67	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_202738	196	441	+	esxN	1-245/285	3/7	84.91	94.38	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_185703	159	401	-	esxN	37-283/285	5/6	84.91	91.13	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_8332	1	247	+	esxH	39-288/291	1/3	84.88	97.6	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_191199	1	249	+	esxH	44-291/291	2/3	84.88	82.8	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_209953	1	254	+	esxM	44-295/297	1/2	84.85	96.85	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_109260	1	256	-	esxM	3-257/297	4/7	84.85	95.37	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_66682	1524	1780	+	esxM	1-255/297	2/8	84.85	95.0	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_138511	1	252	-	esxM	13-264/297	0/0	84.85	82.54	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_114469	1	240	-	esxN	1-240/285	0/0	84.21	90.42	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_114455	1	239	-	esxN	1-240/285	1/1	83.86	93.75	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_17382	866	1114	+	esxM	1-255/297	3/6	83.84	95.29	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_202291	1125	2735	-	eccA3	234-1896/1896	61/96	83.81	87.89	NP_214796	(eccA3) Type VII secretion system protein EccA3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_211779	101	346	-	esxG	39-286/294	1/2	83.67	98.39	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_235095	3	247	+	esxH	28-278/291	5/10	83.51	80.24	NP_214802	(esxH) Type VII secretion system protein ESXH (10 kDa antigen) (CFP-7) (protein TB10.4) [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_179674	836	1087	+	esxM	4-276/297	7/29	83.5	85.92	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_215016	196	971	-	mbtJ	135-910/921	11/16	83.39	94.0	YP_177876	(mbtJ) Putative acetyl hydrolase MbtJ [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_194689	1	240	-	esxA	1-244/288	3/4	83.33	98.36	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_123614	850	1707	+	eccE3	1-836/996	15/34	83.33	92.71	NP_214806	(eccE3) Type VII secretion system transmembrane protein EccE3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_91695	72	318	+	esxM	1-247/297	0/0	83.16	96.36	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_229711	1	1904	+	espK	108-1999/2190	64/154	83.15	87.29	NP_218396	(espK) ESX-1 type VII secretion system EspK [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_41176	1	1467	+	irtB	240-1740/1740	37/78	83.05	89.23	NP_215865	(irtB) Iron-regulated transporter IrtB [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_246634	1	361	+	hspX	22-387/435	4/5	82.99	97.27	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301
k141_23447	1	1202	+	eccB1	155-1382/1443	20/38	82.88	94.81	NP_218386	(eccB1) ESX-1 type VII secretion system protein EccB1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_16286	1	258	-	PE5	1-260/309	5/6	82.85	94.66	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_188150	106	351	+	esxM	16-263/297	2/2	82.83	97.98	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_147527	214	461	+	esxM	1-247/297	2/3	82.83	95.58	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_27803	1	236	-	esxN	1-239/285	1/3	82.81	88.7	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_29447	178	1325	-	mycP3	234-1386/1386	4/9	82.68	98.27	NP_214805	(mycP3) Type VII secretion system membrane-anchored mycosin MycP3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_25178	1	243	+	esxG	51-294/294	1/1	82.65	99.18	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_57867	1	525	+	mbtK	110-633/633	3/3	82.62	97.91	NP_215863	(mbtK) Lysine N-acetyltransferase MbtK [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_116509	1512	2938	-	mbtA	231-1694/1698	40/87	82.57	88.78	NP_216900	(mbtA) Bifunctional enzyme MbtA: salicyl-AMP ligase (SAL-AMP ligase) + salicyl-S-ArCP synthetase [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_121600	1	259	-	PE5	1-255/309	2/4	82.52	98.46	YP_177710	(PE5) PE family protein [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_70024	1	257	-	espC	1-259/312	1/2	82.37	96.14	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_45969	164	1315	-	espB	205-1383/1383	26/55	82.28	94.13	NP_218398	(espB) ESX-1 type VII secretion system EspB [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_28547	308	552	-	esxM	54-297/297	1/1	82.15	97.55	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_8325	38	281	+	esxM	1-246/297	1/2	82.15	96.34	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_60430	1	569	-	ideR	1-571/693	2/2	82.11	97.55	NP_217227	(ideR) Iron-dependent repressor and activator IdeR [IdeR (VF0300)] [Mycobacterium tuberculosis H37Rv]	VF0300

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_164553	193	426	-	esxN	51-285/285	1/1	82.11	93.62	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_53901	768	1005	+	esxN	46-282/285	6/7	82.11	86.31	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_60882	1	357	-	hspX	71-431/435	2/4	82.07	98.34	NP_216547	(hspX) Heat shock protein HspX (alpha-crystallin homolog) (14 kDa antigen) (HSP16.3) [HspX (VF0301)] [Mycobacterium tuberculosis H37Rv]	VF0301
k141_53404	1365	2074	+	erp	1-705/855	11/13	81.99	95.52	NP_218327	(erp) exported repetitive protein precursor Erp [Erp (VF0312)] [Mycobacterium tuberculosis H37Rv]	VF0312
k141_162924	1	1066	+	icl	224-1287/1287	13/20	81.97	94.88	YP_177728	(icl) Isocitrate lyase Icl (isocitrase) (isocitratase) [Isocitrate lyase (VF0253)] [Mycobacterium tuberculosis H37Rv]	VF0253
k141_106903	34	734	+	erp	1-705/855	5/6	81.87	98.3	NP_218327	(erp) exported repetitive protein precursor Erp [Erp (VF0312)] [Mycobacterium tuberculosis H37Rv]	VF0312
k141_65376	1	252	-	esxB	1-248/303	2/4	81.85	96.83	NP_218391	(esxB) 10 kDa culture filtrate antigen EsxB (CFP-10) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_174872	1	1190	+	eccB1	249-1443/1443	15/23	81.84	94.27	NP_218386	(eccB1) ESX-1 type VII secretion system protein EccB1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_203456	239	930	-	lipF	135-834/834	17/28	81.77	94.65	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307
k141_219568	1	1244	+	eccD5	232-1512/1512	24/53	81.75	93.02	NP_216311	(eccD5) ESX-5 type VII secretion system EccD5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_164540	540	772	-	esxN	51-285/285	1/2	81.75	91.06	YP_177838	(esxN) ESX-5 type VII secretion system EsxA (ESAT-6) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_70727	469	713	+	PE35	1-248/300	2/3	81.67	94.76	YP_178021	(PE35) PE family-related protein PE35 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_19492	779	1018	-	esxG	49-290/294	1/2	81.63	86.78	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_244753	233	1076	+	fbpC	1-879/1023	22/53	81.62	90.2	YP_177694	(fbpC) Secreted antigen 85-C FbpC (85C) (antigen 85 complex C) (AG58C) (mycolyl transferase 85C) (fibronectin-binding protein C) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_252984	1431	2874	+	irtB	1-1499/1740	44/105	81.55	89.83	NP_215865	(irtB) Iron-regulated transporter IrtB [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_130537	1212	4787	+	mbtF	1-3578/4386	2/2	81.53	99.72	NP_216895	(mbtF) Peptide synthetase MbtF (peptide synthase) [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_200153	1	242	-	esxM	1-242/297	0/0	81.48	97.52	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_198743	95	549	-	espD	99-555/555	5/8	81.44	97.39	NP_218131	(espD) ESX-1 type VII secretion system EspD [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_25055	1	1825	-	espK	1-1831/2190	35/94	81.32	91.63	NP_218396	(espK) ESX-1 type VII secretion system EspK [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_151637	1	975	+	espA	187-1174/1179	30/47	81.26	92.74	NP_218133	(espA) ESX-1 type VII secretion system EspA [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_198122	997	1230	+	esxA	1-242/288	6/8	81.25	95.04	YP_178023	(esxA) 6 kDa early secretory antigenic target EsxA (ESAT-6) [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_207606	1	1060	-	mbtG	1-1090/1296	26/44	81.25	90.7	NP_216894	(mbtG) Lysine-N-oxygenase MbtG (L-lysine 6-monooxygenase) (lysine N6-hydroxylase) [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299
k141_203530	1	689	-	lipF	1-689/834	13/26	81.06	92.59	NP_218004	(lipF) Probable esterase/lipase LipF [LipF (VF0307)] [Mycobacterium tuberculosis H37Rv]	VF0307

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_153616	117	1611	-	eccA5	332-1833/1833	14/27	81.01	95.44	NP_216314	(eccA5) ESX-5 type VII secretion system AAA+ ATPase EccA5 [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_142980	1	1415	-	irtB	148-1571/1740	16/21	80.98	97.13	NP_215865	(irtB) Iron-regulated transporter IrtB [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_237911	1	721	+	espG3	161-888/888	8/13	80.86	96.85	NP_214803	(espG3) Type VII secretion system protein EspG3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_215175	1558	2079	+	mbtK	3-523/633	13/21	80.73	85.9	NP_215863	(mbtK) Lysine N-acetyltransferase MbtK [mycobactin (IA031)] [Mycobacterium tuberculosis H37Rv]	NA
k141_17186	1	1152	-	eccE1	5-1159/1389	36/69	80.56	90.24	NP_218399	(eccE1) ESX-1 type VII secretion system EccE1 [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_159550	1	1272	-	mbtM	1-1297/1566	32/47	80.52	92.66	NP_215861	(mbtM) Probable fatty acyl-AMP ligase MbtM [FadD33 (VF0308)] [Mycobacterium tuberculosis H37Rv]	VF0308
k141_152838	1	242	-	esxM	1-240/297	2/4	80.47	95.47	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_213104	1	252	-	espC	1-251/312	1/1	80.45	98.41	NP_218132	(espC) ESX-1 type VII secretion system EspC [ESX-1 (VF0315)] [Mycobacterium tuberculosis H37Rv]	VF0315
k141_54086	1	1312	+	eccB3	268-1617/1617	28/62	80.4	91.92	NP_214797	(eccB3) Type VII secretion system protein EccB3 [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_45632	1	239	+	esxG	57-294/294	3/5	80.27	93.78	NP_214801	(esxG) Type VII secretion system protein EsxG [ESX-3 (VF0521)] [Mycobacterium tuberculosis H37Rv]	VF0521
k141_126964	22	1201	-	phoR	272-1458/1458	18/27	80.25	94.74	NP_215272	(phoR) Possible two component system response sensor kinase membrane associated PhoR [PhoP/R (CVF331)] [Mycobacterium tuberculosis H37Rv]	NA

SEQUENCE	START	END	STRAND	GENE	COVERAGE	GAPS	%COVERAGE	%IDENTITY	ACCESSION	PRODUCT	VFID
k141_28134	1	834	+	fbpC	185-1023/1023	20/33	80.16	93.55	YP_177694	(fbpC) Secreted antigen 85-C FbpC (85C) (antigen 85 complex C) (AG58C) (mycolyl transferase 85C) (fibronectin-binding protein C) [Antigen 85 (VF0311)] [Mycobacterium tuberculosis H37Rv]	VF0311
k141_152832	1	238	-	esxM	1-240/297	2/2	80.13	95.83	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_223049	1	243	+	esxM	49-287/297	4/6	80.13	93.44	NA	(esxM) ESX-5 type VII secretion system EsxB (CFP-10) homolog [ESX-5 (VF0402)] [Mycobacterium tuberculosis H37Rv]	VF0402
k141_51709	1245	1417	+	mbtH	43-216/216	1/1	80.09	97.7	NP_216893	(mbtH) putative protein MbtH [Mycobactin (VF0299)] [Mycobacterium tuberculosis H37Rv]	VF0299

SIGNIFICANT VIRULENCE FACTORS - DESCRIPTION

	VF_Name	VF_FullName	Bacteria	VFCID	VFcategory	Characteristics	Structure	Function	Mechanism
7	LipF	NA	Mycobacterium tuberculosis	VFC0272	Nutritional/Metabolic factor	NA	NA	Annotated as a lipase/esterase, dramatically upregulated at low pH, a possible function of lipF could be to hydrolyze toxic fatty acids present in caseous necrotic debris during tuberculosis or it may modify the external cell wall as an adaptive response to acid damage.	NA
9	MgtC	NA	Mycobacterium tuberculosis	VFC0272	Nutritional/Metabolic factor	Homology to MgtC of Salmonella enterica, which is essential for the survival of S.enterica within macrophages.	NA	Magnesium acquisition.	NA
2	ESX-5	ESAT-like secretion system	Mycobacterium tuberculosis	VFC0086	Effector delivery system	NA	NA	Essential for the secretion of the ESX-5-encoded ESAT-6-family member EsxN by M. marinum; necessary and sufficient to produce a functional secretion machinery for PPE41 in M. smegmatis.	NA
1	ESX-3	NA	Mycobacterium tuberculosis	VFC0086	Effector delivery system	NA	NA	Required for siderophore-mediated iron acquisition.	NA

	VF_Name	VF_FullName	Bacteria	VFCID	VFcategory	Characteristics	Structure	Function	Mechanism
1	Antigen 85	NA	Mycobacterium tuberculosis	VFC0001	Adherence	Three homologous proteins: antigen 85A, 85B and 85C, also known as fibronectin-binding protein (Fbp).	Containing a carboxylesterase domain bearing the highly-conserved consensus sequence GXXSXXG; PDB code: 1DQZ.	Mycolyltransferase activity that transfer long-chain mycolic acids to trehalose derivatives (i.e., cord factor, a toxic glycolipid consisting of two mycolic acid molecules linked to trehalose by the hydroxyl groups of carbons 6 and 6' --> trehalose 6,6'-dimycolate (TDM), is an important immunomodulatory component of the cell wall), pivotal for the biosynthesis of the mycobacterial cell wall and for the survival of mycobacteria; binds the cell matrix protein fibronectin thereby enhancing the uptake of mycobacteria macrophages via complement-mediated phagocytosis.	NA
9	Mycobactin	NA	Mycobacterium tuberculosis	VFC0272	Nutritional/Metabolic factor	Mycobacteria produce two classes of siderophores, mycobactins and the exochelins. Pathogenic mycobacteria solely produce mycobactins, whereas saprophytic mycobacteria such as M. smegmatis and Mycobacterium neoarum produce both mycobactins and exochelins; Mycobactins are salicylate containing siderophores, and exochelins are peptidic molecules; Mycobactins are found in two forms that differ in the length of an alkyl substitution and hence in polarity and solubility. The less polar form remains cell associated (mycobactin), whereas the more polar one (carboxymycobactin) is secreted into the medium.	NA	Cell-association mycobactin participates in iron internalization and/or to serve as a temporary iron-holding molecule to prevent sudden influx of excess iron if the metal suddenly becomes available after a period of iron limitation.	NA

	VF_Name	VF_FullName	Bacteria	VFCID	VFcategory	Characteristics	Structure	Function	Mechanis
	NA	NA	NA	NA	NA	NA	NA	NA	NA
3	Isocitrate lyase	NA	Mycobacterium tuberculosis	VFC0346	Others	NA	PDB code: 1F61.	Required for persistent infection.	Isocitrate ly the initial e in the glyox shunt, a secondary metabolic pathway th allows bact utilize fatty as carbon energy sou when the availability primary ca sources is limiting; Iso lyase, in combinatio malate syn catalyzes t conversion isocitrate to malate, a reaction th allows maintenan the TCA cy and synthe carbohydra from fatty a
5	ESX-1	ESAT-6 secretion system-1	Mycobacterium tuberculosis	VFC0086	Effector delivery system	The ESX-1 secretion system is encoded by two different loci, the ESX-1 locus and the EspA operon. The exact number of components that are involved in ESX-1 secretion is still debated and seems to vary between different mycobacterial species; All the known ESX-1 substrates, Rv3616c(EspA), Rv3615c, ESAT-6 and CFP-10 are mutually dependent on each other for secretion.	NA	ESX-1 is required for survival of mycobacteria in vivo; required for (initiation of) granuloma formation; essential for escape from phagosomes into cytosol; essential for cell lysis and cell-to-cell spreading; required for phagosome maturation; essential for induction of immunosuppressive type I interferons and represses conjugation in donor cells (M. smegmatis).	NA
3	NA	NA	NA	NA	NA	NA	NA	NA	NA
0	IdeR	Iron-dependent regulator	Mycobacterium tuberculosis	VFC0301	Regulation	A DNA binding protein interacts with a conserved DNA sequence requires the binding of Fe2+ or related divalent cations.	NA	Repressor of iron acquisition genes, activator of iron storage genes and a positive regulator of oxidative stress responses.	NA

	VF_Name	VF_FullName	Bacteria	VFCID	VFcategory	Characteristics	Structure	Function	Mechanis
1	HspX	16-kDa alpha-crystallin-like protein	Mycobacterium tuberculosis	VFC0282	Stress survival	NA	NA	ATP-independent chaperone, associated with stationary phase, induced under RNI stress and is an immunodominant antigen that is necessary for survival in macrophages. This protein can stabilize cell structure during long-term survival and permit the bacilli to survive within the low-oxygen environment of the granuloma.	NA
7	RelA	NA	Mycobacterium tuberculosis	VFC0301	Regulation	NA	NA	Associated with stationary phase adaptation and long-term survival.	Produce pp (guanosine 5'-diphospl 3'-diphospl an 'alarmo suppresse synthesis c stable RNA induces degradativ pathways a modulates expression genes invo DNA replic
2	Erp	Exported repetitive protein	Mycobacterium tuberculosis	VFC0346	Others	Surface-located protein;No homologue of Erp has been found in other bacterial species, making Erp a mycobacterial signature.	Three domains: a highly conserved N-terminal domain including a signal sequence, a central variable region containing repeats based on the motif (PA/G)LTS, and a conserved C-terminal domain rich in proline and alanine.	Play roles in the survival and multiplication of mycobacteria in macrophages.	Interacting Rv2212, ar adenylyl cy to enhance 12-mediate cyclic AMP (cAMP) production
6	PhoP	NA	Mycobacterium tuberculosis	VFC0301	Regulation	Similar to the PhoP response regulator of S. enteric serovar Typhimurium that senses Mg2+ starvation and controls the expression of virulence genes.	NA	Senses Mg2+ starvation, controls expression of genes involved in surface remodeling and adaptation to intracellular growth.	NA

	VF_Name	VF_FullName	Bacteria	VFCID	VFcategory	Characteristics	Structure	Function	Mechanisr
3	HbhA	NA	Mycobacterium tuberculosis	VFC0001	Adherence	Surface-located protein.	NA	A multifunctional protein involved in adherence of Mycobacterium tuberculosis to non-phagocytic cells and in the formation of intracytosolic lipid inclusions.	NA

OTHER VIRULENCE FACTORS

No other virulence factors detected.

ANALYSIS PIPELINE SPECIFICATIONS

Task	Tool	Version
resistance_prediction	tb-profiler	6.6.5
mapping	minimap2	2.30
variant_calling	bcftools	1.22
long_variant_calling	delly	1.3.3
depth_calculation	samtools	1.22.1
virulence_prediction	abricate	1.0.1
spoligotyping	SpoTyping	2.1
assembly	MEGAHIT	1.2.9

Database: tbdb (commit: 0a909094, date: Tue Aug 5 10:43:00 2025 +0100)

TARGET REGIONS

Target genes (73 regions):

dnaA, gyrB, gyrA, Rv0010c, fgd1, mshA, ccsA, Rv0565c

hadA, nusG, mmaA3, rpoB, rpoC, mmpL5, mmpS5, mmpR5

rpsL, rplC, Rv1129c, fbiC, sigE, Rv1258c, embR, atpE

rrs, rrl, inhA, rpsA, tsnR, tlyA, bacA, ndh

katG, PPE35, Rv1979c, pncA, kasA, eis, ahpC, folC

Rv2477c, pepQ, ribD, Rv2680, Rv2681, Rv2752c, thyX, thyA

ald, fbiD, Rv3083, whiB7, Rv3236c, lpqB, mtrB, mtrA

fbiA, fbiB, alr, rpoA, ddn, clpC1, panD, glpK

embC, embA, embB, aftB, ubiA, ethA, ethR, whiB6

gid

INFO

General

Sample ID:	ERR9030255
Species:	Mycobacterium tuberculosis
Analysis date:	2026-02-27T19:53:49.556287
TB-Profiler version:	6.6.5
Database:	tbdb (0a909094)
Reads mapped:	171808
% reads mapped:	100.0
Median depth:	118.0

Lineage

Main lineage:	lineage1
Sub lineage:	lineage1.1.2
Family:	Indo-Oceanic
RD:	RD239
Spoligotype octal:	477403377413771
Spoligotype binary:	1001111111000000110111111111000010111111111

Resistance

Drug resistance type:	Sensitive
Number of resistance variants:	0
Number of other variants:	71

Disclaimer

This report is generated for research purposes. Drug resistance predictions are based on the WHO catalogue of mutations and should be confirmed by phenotypic drug susceptibility testing (DST) before clinical decisions are made. The analysis uses the TB-Profiler bioinformatics pipeline with the tbdb mutation database. Results should be interpreted by qualified personnel in the context of clinical and epidemiological data.

CITATIONS

Phelan JE, O'Sullivan DM, Machado D, et al. Integrating informatics tools and portable sequencing technology for rapid detection of resistance among tuberculosis bacteria. Genome Medicine 11, 41. 2019.

https://github.com/tseemann/abricate

Zhou S, Liu B, Zheng D, Chen L, Yang J. VFDB 2025: an integrated resource for exploring anti-virulence compounds. Nucleic Acids Res. 2025.

Xia E, Teo YY, Ong RT. SpoTyping: fast and accurate in silico Mycobacterium spoligotyping from sequence reads. Genome Med. 2016;8(1):19.

Li H. Minimap2: pairwise alignment for nucleotide sequences. Bioinformatics. 2018;34(18):3094-3100.

Li D, Liu CM, Luo R, Sadakane K, Lam TW. MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de Bruijn graph. Bioinformatics. 2015;31(10):1674-1676.